

Unit 7 - Trigonometric Ratios	
Lesson	Page Number
7.1 The Pythagorean Theorem	369
7.2 Defining Sine and Cosine	374
7.3 Defining Tangent	381
7.4 Trigonometric Ratios on the Coordinate Plane	387
7.5 Using Trigonometric Ratios	393
7.6 Special Right Triangles	400
7.7 The Unit Circle and Its Coordinates	407
7.8 Using Trigonometric Ratios and the Pythagorean Theorem	416
7.9 Using Trigonometry in the Real World	421
7.H1 Law of Sines	427
7.H2 Law of Cosines	433
7.H3 Area of a Triangle Using the Sine Ratio	439

The following B.E.S.T Standards are covered in this unit:

MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles.

- *Clarification 1: Instruction includes using the Pythagorean Theorem and using similar triangles to demonstrate that trigonometric ratios stay the same for similar right triangles.*
- *Clarification 2: Within the Geometry course, instruction includes using the coordinate plane to make connections to the unit circle.*
- *Clarification 3: Within the Geometry course, trigonometric ratios are limited to sine, cosine and tangent.*

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.

- *Clarification 1: Instruction includes procedural fluency with the relationships of side lengths in special right triangles having angle measures of $30^\circ - 60^\circ - 90^\circ$ and $45^\circ - 45^\circ - 90^\circ$.*

MA.912.T.1.3 (Honors only) Apply the Law of Sines and the Law of Cosines to solve mathematical and real-world problems involving triangles.

MA.912.T.1.4 (Honors only) Solve mathematical problems involving finding the area of a triangle given two sides and the included angle.

- *Clarification 1: Problems include right triangles, heights inside of a triangle and heights outside of a triangle.*

Unit 7, Lesson 1: The Pythagorean Theorem



Benchmarks

MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles.

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.



Learning Target

- ☐ I can find the missing side of a right triangle using the Pythagorean Theorem.



7.1.1 Warm-Up: Sales Strategist

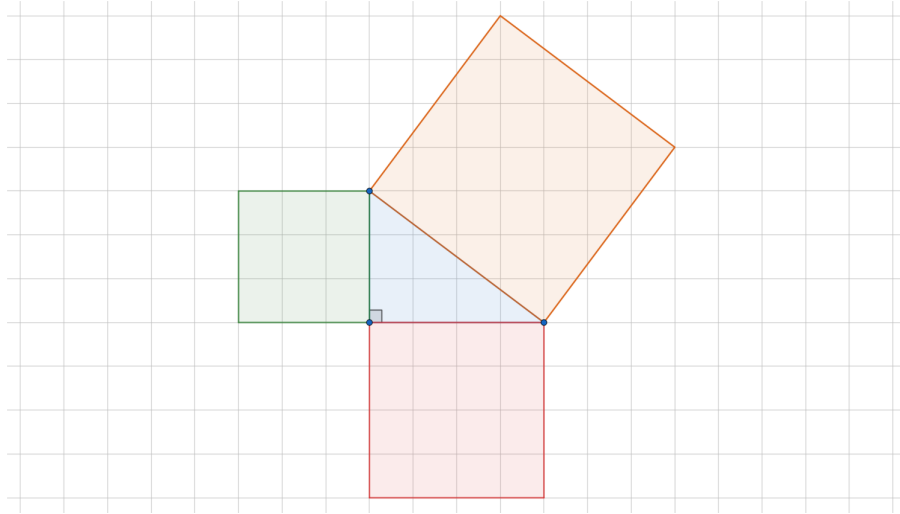
Kishan Patel says that he was never really a good student in high school. He graduated with a 2.9 GPA. All of those challenges prepared him for business. Business is about failing and doing better. Every time you fail, you have to improve in some way to get to the next level.

1. Based on the statement above, how do you think Kishan views failure?
2. When you fail at something, describe how you typically react.

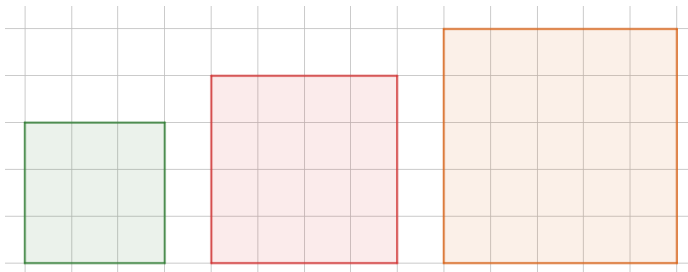


7.1.2 Collaboration: Pythagorean Theorem

A right triangle is shown with three squares.

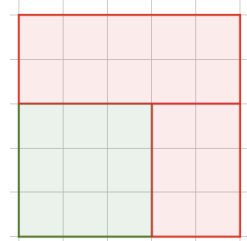


1. The three squares from the diagram are shown. Complete the table with the area of each square.



Side Length (units)	Area of Square (square units)
3	
4	
5	

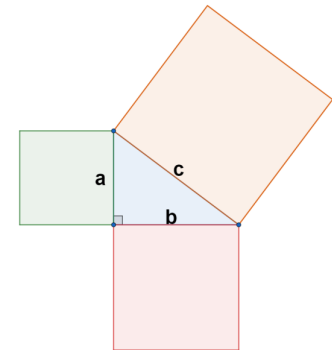
2. Explain how the diagram shows that the area of the green square plus the area of the red square is equal to the area of the orange square.



3. Complete the relationship of the areas by filling in the box.

$$3^2 + 4^2 \square 5^2$$

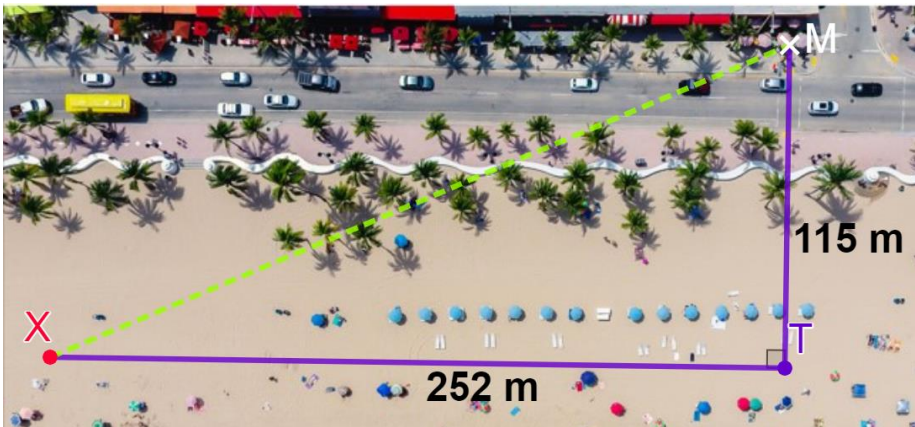
4. Use the diagram to write an explanation of the Pythagorean Theorem, $a^2 + b^2 = c^2$.



5. Iverson and Jake are at the beach and plan to meet a friend for lunch.



Iverson decided to walk directly to the restaurant, while Jake decides to take a route that will allow him to cross the street using the crosswalk. The diagram shows the two routes. Jake walks horizontally 252 meters (m) and then vertically 115 m.



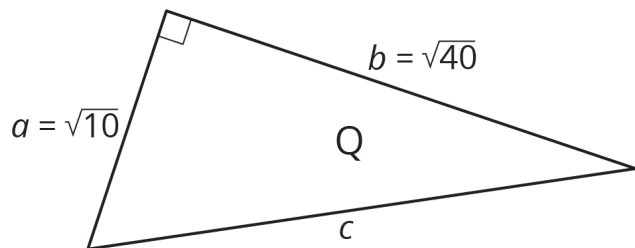
- a. Complete the table.

	Iverson	Jake
Distance Walked		

- b. Discuss with your partner the advantages and disadvantages of each person’s walk to the restaurant. Summarize your discussion in the table.

Person	Advantages	Disadvantages
Iverson		
Jake		

6. A right triangle is shown with side lengths $\sqrt{10}$ and $\sqrt{40}$ units.



Find the length of the hypotenuse using the Pythagorean Theorem. If the measure of the side length is an irrational value, write an exact solution and an approximate solution.

- a. State Pythagorean Theorem

$$a^2 + b^2 = c^2$$

- b. Substitute

$$\underline{\hspace{1cm}}^2 + \underline{\hspace{1cm}}^2 = c^2$$

- c. Solve for the missing variable.

$$\underline{\hspace{1cm}} + \underline{\hspace{1cm}} = c^2$$

$$\underline{\hspace{1cm}} = c^2$$

- d. Solutions

Exact solution:

$$c = \sqrt{\hspace{1cm}}$$

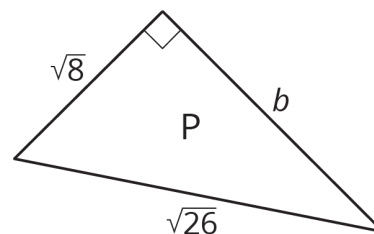
Approximate answer:

$$c \approx \underline{\hspace{1cm}}$$

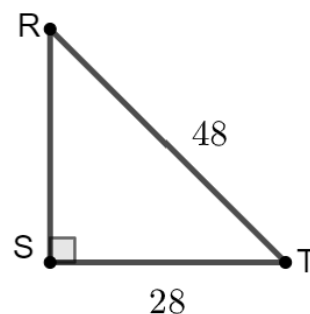


7.1.3 Guided Practice: Pythagorean Theorem

1. Find the exact measure of b in the diagram.

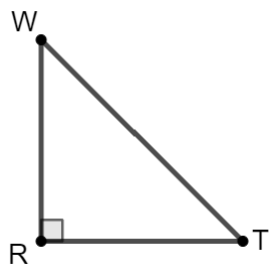


2. $\triangle RST$ is shown, where $ST = 28$ and $RT = 48$.



Find the length of $\overline{SR}$ to the nearest thousandth.

3. $\triangle WRT$ is shown, where $WT = 18.4$ and $RT = 12.1$.



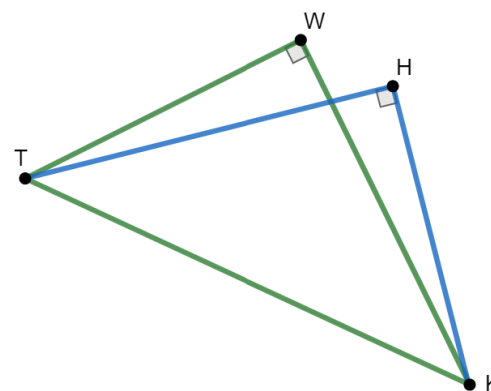
Find the length, to the nearest thousandth, of $\overline{WR}$.



7.1.4 Your Turn!

1. In $\triangle HMX$, $m\angle MXH = 90^\circ$, $MH = 12$, and $MX = 7$. Determine the length of $\overline{XH}$. Round to the nearest thousandth.

2. $\triangle TWK$ and $\triangle THK$ are shown, where $TW = \sqrt{45}$, $TH = \sqrt{68}$, and $TK = \sqrt{113}$. Find the length of $\overline{WK}$ and $\overline{HK}$. Round to the nearest thousandth.





7.1.5 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about the Pythagorean Theorem.

Unit 7, Lesson 2: Defining Sine and Cosine



Benchmarks

MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles.

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.



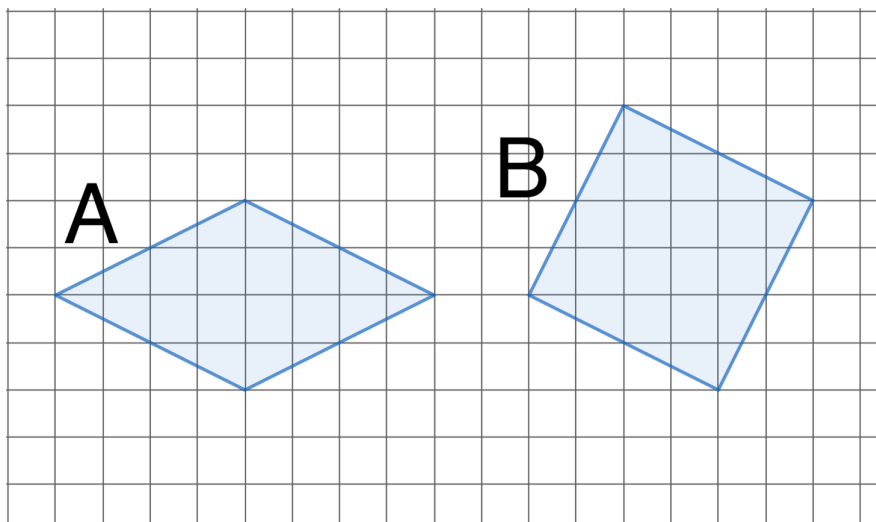
Learning Target

- ☐ I can define sine and cosine using similar triangles.



7.2.1 Warm-Up: Which Region Is Larger?

- Two grids are shown with regions shaded.



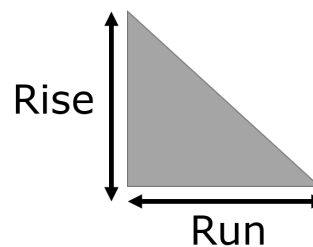
- Which shaded region is larger?
- Explain your reasoning.



7.2.2 Exploration: Similar Right Triangles

- Using the space below, create a right triangle that has one angle with a measure of 60° . Be accurate with measurements.

- Measure the rise, run, and slope of your triangle in centimeters. Round to the nearest tenth.



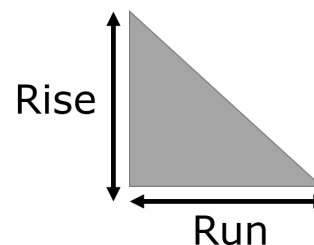
Rise	Run	Slope $\left(\frac{\text{rise}}{\text{run}}\right)$

3. Record the slopes from two of your classmates. *You are the only person who should write in the first two columns of the table.* Have the person initial stating that your slope is correct.

Classmate	Slope $\left(\frac{\text{rise}}{\text{run}}\right)$	Initials

- a. What do you notice about the slopes?
- b. What do you wonder?
4. Draw a right triangle that has an angle of 45° . Be accurate with measurements.

5. Measure the rise and run of your triangle in centimeters. Round to the nearest tenth.



Rise	Run	Slope $\left(\frac{\text{rise}}{\text{run}}\right)$

6. Record the slopes from two of your classmates. *You are the only person who should write in the second column of the table.* Have the person initial stating that your slope is correct.

Classmate	Slope $\left(\frac{\text{rise}}{\text{run}}\right)$	Initials

- a. What do you notice about the slopes?
- b. What do you wonder?

7. Ask two classmates for the “Notice and Wonder” what they wrote for both triangles. Record their responses below. *You are the only person who should write in the first three columns of the table.* Have the person initial that it is written correctly.

Classmate	Notice	Wonder	Initials

8. Write a conjecture for each of the right triangles that were drawn.

Angle Measures	Conjecture
$30^\circ - 60^\circ - 90^\circ$	
$45^\circ - 45^\circ - 90^\circ$	

9. The triangles drawn by you and your classmates during the Exploration vary. Using either the triangles explored in Questions 1-3 or Questions 4-6, explain why these triangles are similar.



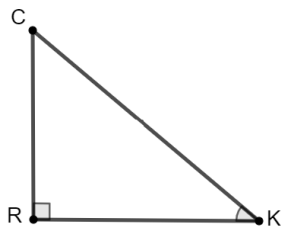
Right triangles with one other congruent angle are similar because of AA similarity.



7.2.3 Collaboration: Introduction to Trigonometric Ratios

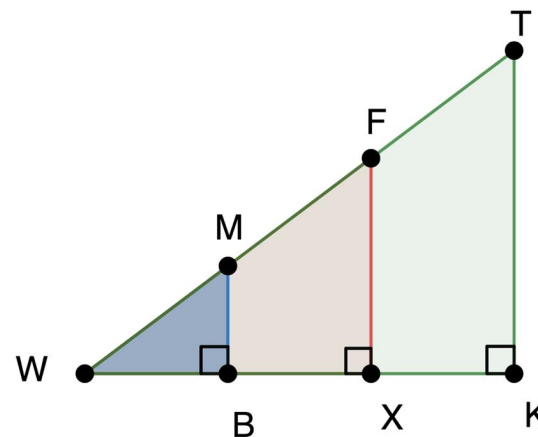
Work with your partner to complete the following.

1. $\triangle CRK$ is shown, where $m\angle CRK = 90^\circ$ and $\angle CKR$ is marked as a reference angle.



- Draw an arrow from $\angle CKR$ to side $\overline{CR}$. Label that side with an o for opposite.
- Label side $\overline{RK}$ with an a for adjacent.
- Label the hypotenuse, $\overline{CK}$, with an h for hypotenuse.

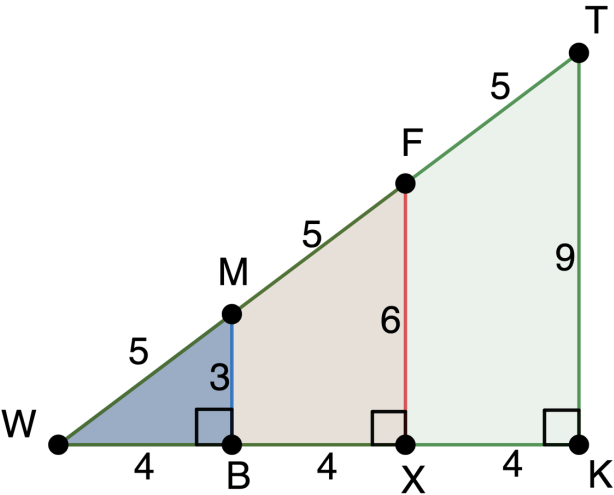
2. The diagram shows three right triangles that share the reference $\angle W$. Explain why $\triangle WMB$, $\triangle WFX$, and $\triangle WTK$ are similar.



3. Complete the table.

Triangle	Leg Opposite $\angle W$	Leg Adjacent to $\angle W$	Hypotenuse
$\triangle WMB$			
$\triangle WFX$			
$\triangle WTK$			

4. The diagram shows measurements for the three triangles.



Complete the table using $\angle W$ as the reference angle.

Triangle	$\frac{\text{opposite}}{\text{hypotenuse}}$	$\frac{\text{adjacent}}{\text{hypotenuse}}$	$\frac{\text{opposite}}{\text{adjacent}}$
$\triangle WMB$			
$\triangle WFX$			
$\triangle WTK$			



7.2.4 Bring It Together: Trigonometric Ratios

Since $\triangle WMB$, $\triangle WFX$, and $\triangle WTK$ are similar, the $\frac{\text{opposite}}{\text{hypotenuse}}$ ratios are equivalent to each other. This is also true for the ratios $\frac{\text{adjacent}}{\text{hypotenuse}}$ and $\frac{\text{opposite}}{\text{adjacent}}$. For this reason, it is possible to define these ratios in any right triangle.

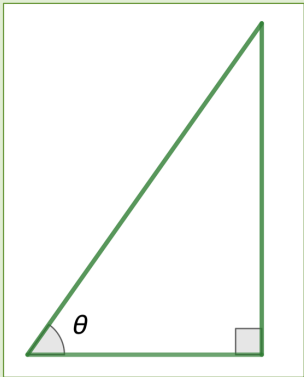
The ratios use the reference angle theta (θ).

Trigonometric Ratios

$$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$$

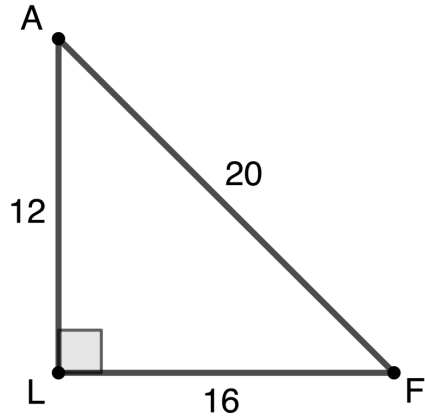
$$\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}}$$





7.2.5 Guided Practice: Trigonometric Ratios

1. $\triangle FLA$ is shown. $m\angle ALF = 90^\circ$, $AL = 12$, $LF = 16$, and $AF = 20$.

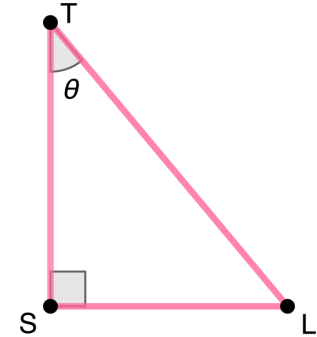


Complete the table using each angle as the reference angle.

Angle	$\frac{\text{opposite}}{\text{hypotenuse}}$	$\frac{\text{adjacent}}{\text{hypotenuse}}$	$\frac{\text{opposite}}{\text{adjacent}}$
$\angle F$			
$\angle A$			

2. $\triangle TLS$ is shown where $LS = \sqrt{35}$ and $TS = \sqrt{42}$.

- a. Determine the length of $\overline{TL}$.



- b. Complete the table for the reference angle θ .

Angle	$\frac{\text{opposite}}{\text{hypotenuse}}$	$\frac{\text{adjacent}}{\text{hypotenuse}}$	$\frac{\text{opposite}}{\text{adjacent}}$
θ			



7.2.6 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to explain what was missed in today's lesson.

Unit 7, Lesson 3: Defining Tangent



Benchmarks

MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles.

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.



Learning Target

- ☐ I can use the ratios of sine and cosine to define the tangent ratio.



7.3.1 Warm-Up: Notice and Wonder



1. What do you wonder?

2. What do you notice?



7.3.2 Exploration: Relating Sine, Cosine and Tangent

1. The table gives the sine and cosine values as decimals for different reference angle measures. Using a calculator and the values in the table, find the ratio of $\frac{\text{sine}}{\text{cosine}}$. Round to the nearest thousandth.

Angle Measure	Sine	Cosine	$\frac{\text{Sine}}{\text{Cosine}}$
10°	0.174	0.985	
20°	0.342	0.940	
30°	0.500	0.866	
40°	0.643	0.766	
50°	0.766	0.643	
60°	0.866	0.500	
70°	0.940	0.342	
80°	0.985	0.174	

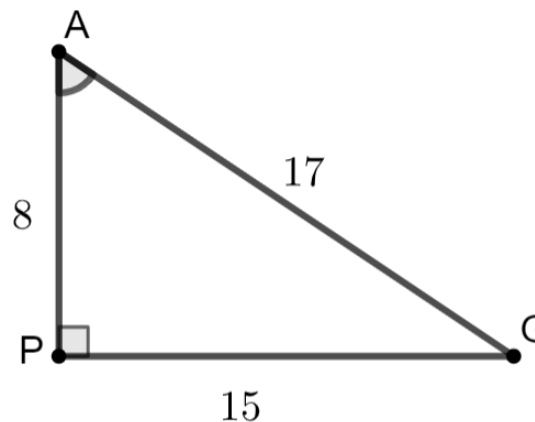
2. Use a calculator to find the tangent ($\tan$) of each reference angle.

Angle Measure	Tan
10°	
20°	
30°	
40°	
50°	
60°	
70°	
80°	

3. Write down one notice statement and one wonder statement about the values of $\frac{\text{sine}}{\text{cosine}}$ from the first table and the values of $\tan$ from the second table.

Notice	Wonder

4. $\triangle APG$ is shown, where $m\angle APG = 90^\circ$, $AP = 8$, $PG = 15$, and $AG = 17$.



- a. Using $\angle PAG$ as the reference angle, complete the table.

Sine $\left(\frac{\text{opposite}}{\text{hypotenuse}}\right)$	Cosine $\left(\frac{\text{adjacent}}{\text{hypotenuse}}\right)$	Tangent $\left(\frac{\text{opposite}}{\text{adjacent}}\right)$

- b. Multiply the sine and the cosine ratios from the table.

$$\underline{\hspace{2cm}} \cdot \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

- c. Rewrite $\text{sine} \cdot \text{cosine}$ using the ratios of sine and cosine, $\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$ and $\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$. If possible, simplify.

- d. Divide the sine ratio by the cosine ratio from the table.

$$\underline{\hspace{2cm}} \div \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

- e. Rewrite $\frac{\text{sine}}{\text{cosine}}$ using the ratios of sine and cosine, $\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$ and $\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$. The resulting ratio is a complex fraction that can be rewritten or simplified. Work with your partner to write the simplified form.

- f. Describe how the ratios of sine and cosine can be used to find the tangent ratio.

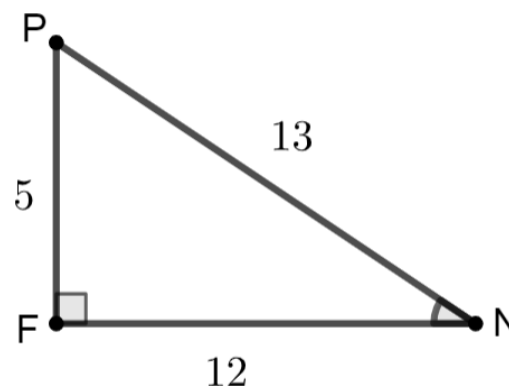


7.3.3 Guided Instruction: Determining Trigonometric Ratios



The tangent ratio of a reference angle can be found using the ratio, $\frac{\text{sine}}{\text{cosine}}$.

1. $\triangle PFN$ is shown, where $m\angle PFN = 90^\circ$, $PF = 5$, $FN = 12$, and $PN = 13$.



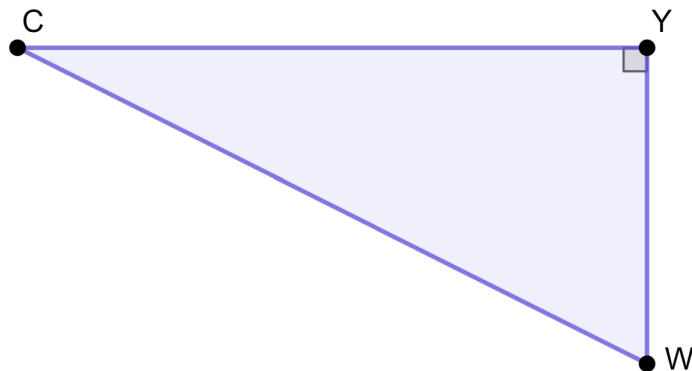
- a. Using $\angle FNP$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent

b. Which method, $\frac{\text{sine}}{\text{cosine}}$ or $\frac{\text{opposite}}{\text{adjacent}}$, did you use to find the tangent ratio?

c. Use the method for finding tangent that you did not use to check the ratio.

2. $\triangle CYW$ is shown, where $CY = \sqrt{98}$ and $YW = \sqrt{28}$.



a. Determine the length of $\overline{CW}$.

b. Using $\angle WCY$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent

c. Using $\angle YWC$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent

d. What do you notice about the ratios for $\angle WCY$ and the ratios for $\angle YWC$?



7.3.4 Guided Practice: Trigonometric Ratios

1. $\triangle YNB$, $m\angle NBY = 90^\circ$, $BN = 6\sqrt{2}$, and $YN = 4\sqrt{17}$.

a. Determine the length of $\overline{YB}$.

b. Using $\angle BNY$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent

2. $\triangle GMK$, $m\angle GKM = 90^\circ$, $GM = 3\sqrt{10}$, and $GK = 3\sqrt{3}$.

a. Determine the length of $\overline{KM}$.

b. Using $\angle KGM$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent



7.3.5 Wrap-Up: Cool Down

1. Write a short note to a friend who missed today's class summarizing what you learned.

Unit 7, Lesson 4: Trigonometric Ratios on the Coordinate Plane



Benchmarks

MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles.

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.



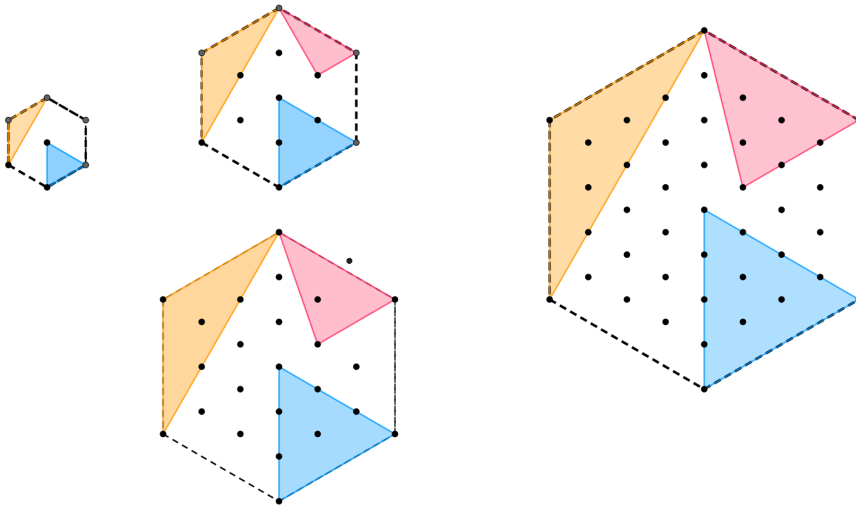
Learning Target

- ☐ I can determine trigonometric ratios using the coordinate plane.



7.4.1 Warm-Up: Patterns

Four hexagons are shown.



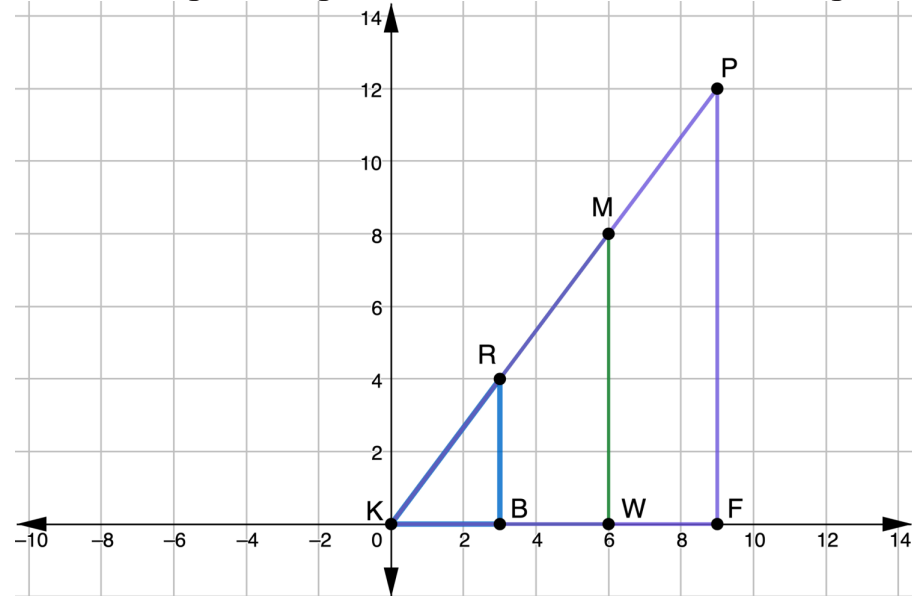
1. Describe the pattern of each triangle.

Triangle	Pattern Description
Yellow	
Red	
Blue	



7.4.2 Collaboration: Defining Trigonometric Ratios

1. Three right triangles are shown on the coordinate grid.



a. Discuss with your partner how to justify that ΔKBR , ΔKWM , and ΔKFP are similar. Summarize your discussion.

b. Complete the table.

Segment	Length	Segment	Length
$\overline{KB}$		$\overline{WM}$	
$\overline{BR}$		$\overline{KF}$	
$\overline{KW}$		$\overline{FP}$	

- c. Use the lengths from the table to determine each. Show your work.

Segment	Length
$\overline{KR}$	
$\overline{KM}$	
$\overline{KP}$	

- d. Using $\angle K$ as the reference angle, find the three ratios for each triangle.

Triangle	Cosine	Sine	Tangent
$\triangle RKB$			
$\triangle MKW$			
$\triangle PKF$			

2. $\triangle HWB$ is graphed on the coordinate grid, where $m\angle WBH = 90^\circ$. The coordinates of the vertices are $H(-4, 5)$, $W(5, -3)$, and $B(-4, -3)$.

- a. Discuss with your partner how to use the coordinates to determine the trigonometric ratios. Summarize your discussion.

- b. Complete the table. Show your work.

Segment	Length
$\overline{WB}$	
$\overline{BH}$	
$\overline{HW}$	

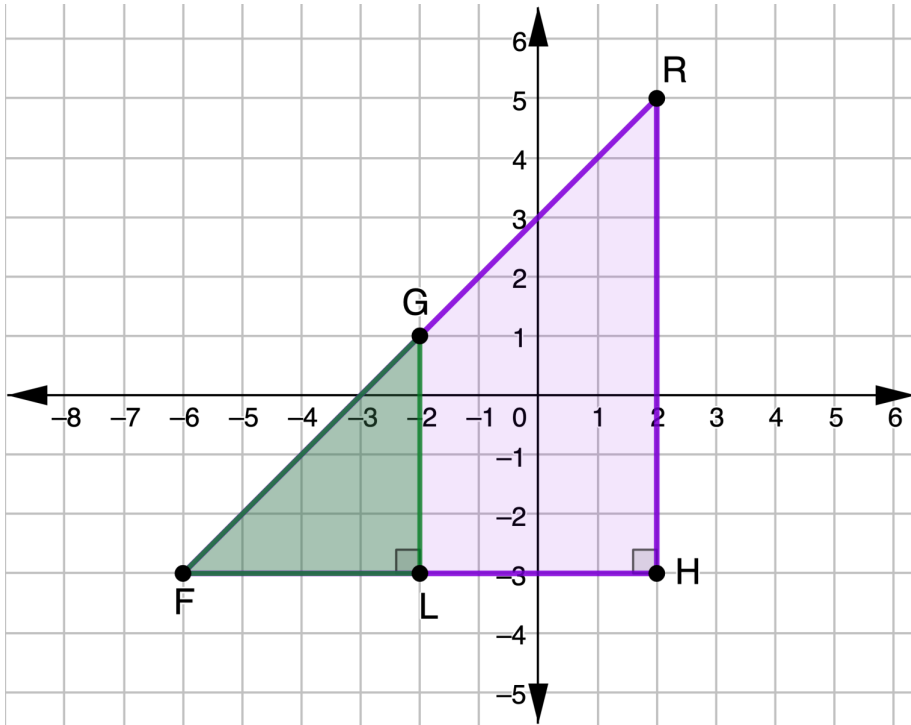
- c. Using $\angle WHB$ as the reference angle, find the three ratios.

Cosine	Sine	Tangent



7.4.3 Guided Instruction: Using Trigonometric Ratios

1. $\triangle FLG$ and $\triangle FHR$ shown on the coordinate grid are isosceles right triangles.



- a. Determine the measure of each angle.

Angle	Measure
$\angle LFG$	
$\angle FGL$	
$\angle FRH$	

- b. Using $\angle F$ as the reference angle, find the three ratios for each triangle.

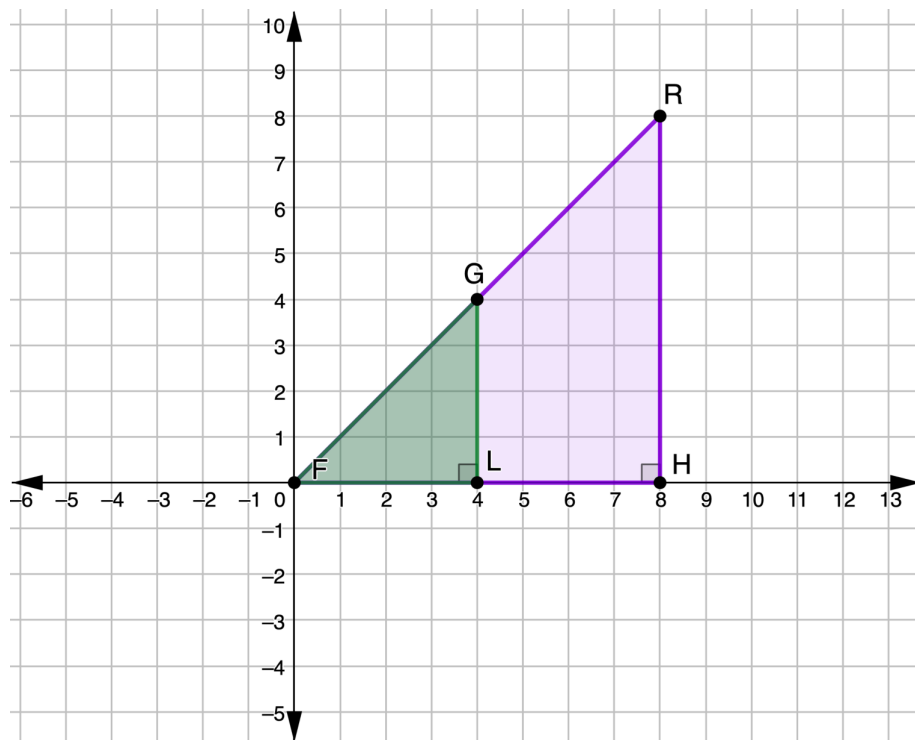
Triangle	Cosine	Sine	Tangent
$\triangle FLG$			
$\triangle FHR$			

- c. Use your calculator to find each. Round to the nearest thousandth, if necessary.

Angle	Cosine	Sine	Tangent
45°			

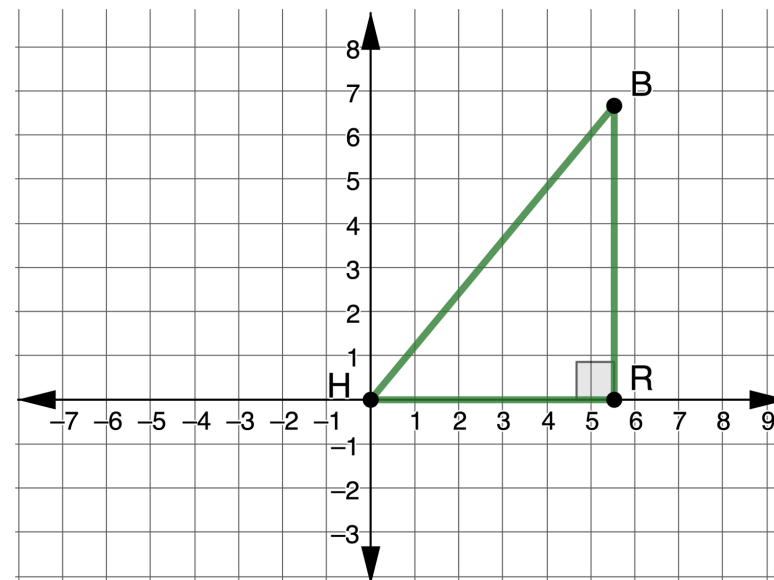
- d. How are the ratios you found using $\angle F$ as the reference angle related to the values you found for the ratios of a 45° angle?

- e. $\triangle FLG$ and $\triangle FHR$ were moved on the coordinate grid so that F is at the origin.



Work with your partner to determine whether or not the values of the trigonometric ratios will change because the triangle moved. Summarize your findings.

2. $\triangle HBR$ is shown on the coordinate grid, where $m\angle HRB = 90^\circ$, $m\angle RHB = 50^\circ$, $HR = 5.567$, and $HB = 8.66$.



- a. Explain why the ratio $\cos(50) \approx \frac{5.567}{8.66}$ is true.

- b. Find each value using a calculator. Round to the nearest thousandth.

$$\cos(50) = \frac{5.567}{8.66} =$$

c. Discuss with your partner why the approximation symbol was used in $\cos(50^\circ) \approx \frac{5.567}{8.66}$. Summarize your thoughts below.

d. Work with your partner to find $\sin(H)$. Round to the nearest thousandth.

e. Use your calculator to find $\sin(50^\circ)$.

f. Explain why $\sin(50^\circ) \approx \sin(H)$.

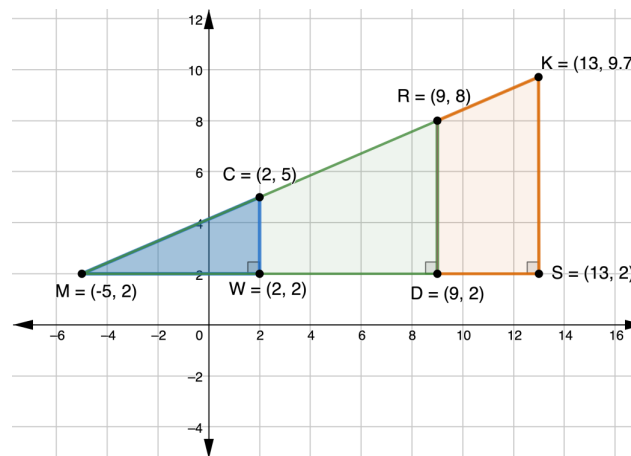
g. Work with your partner to write the coordinates of each vertex.

Vertex	Coordinates
R	
B	



7.4.4 Guided Practice: Trigonometric Ratios and Similar Triangles

1. $\triangle CMW$, $\triangle RMD$, and $\triangle KMS$ are shown on the coordinate grid.



Using $\angle M$ as the reference angle, find the three ratios for each triangle. Write the values as a decimal rounded to the nearest thousandth.

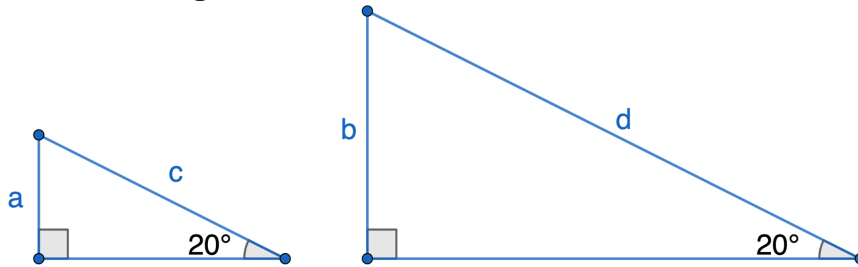
Triangle	Cosine	Sine	Tangent
$\triangle CMW$			
$\triangle RMD$			
$\triangle KMS$			

2. Yasmeen said she only found the ratios for $\triangle CMW$ then applied them to the other triangles. Explain using mathematics why she can do that.



7.4.5 Wrap-Up: Ticket Out the Door

1. Two triangles are shown.



- a. Describe how the ratios $\frac{a}{c}$ and $\frac{b}{d}$ are related.
- b. What mathematical relationship(s) did you use to complete your description?

Unit 7, Lesson 5: Using Trigonometric Ratios



Benchmark

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.



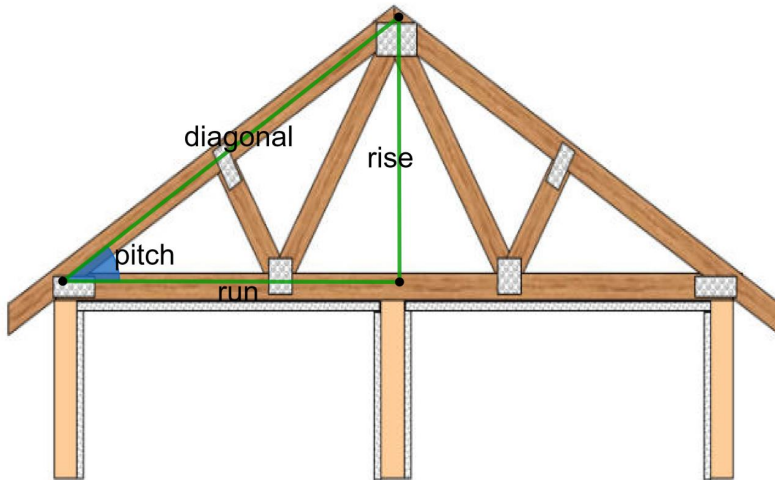
Learning Targets

- ☐ I can use trigonometric ratios to find the side of a right triangle.
- ☐ I can use trigonometric ratios to find an angle of a right triangle.



7.5.1 Warm-Up: Notice and Wonder

In carpentry, right triangles are commonly used in constructing roofs. A roof's pitch is the steepness of the roof. It can be determined by the ratio between the two legs of the right triangle.

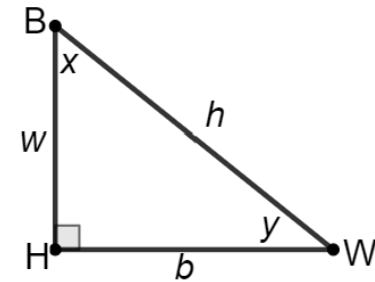


1. What do you notice or wonder about the use of right triangles in construction?



7.5.2 Collaboration: Using Trigonometric Ratios

1. $\triangle BWH$ is shown, where $h = 12.8$, $w = 8$, $x = 51.3^\circ$, and $y = 38.7^\circ$.



- a. Select all of the equations that can be used to find the value of b .

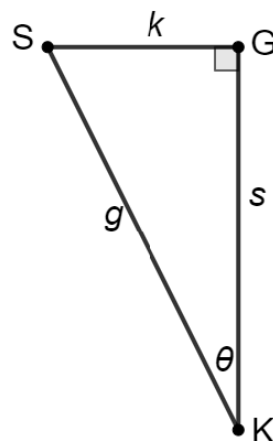
☐ $\cos(51.3^\circ) = \frac{8}{b}$ ☐ $\sin(51.3^\circ) = \frac{b}{12.8}$ ☐ $\tan(51.3^\circ) = \frac{b}{12.8}$

☐ $\cos(38.7^\circ) = \frac{b}{12.8}$ ☐ $\sin(38.7^\circ) = \frac{b}{12.8}$ ☐ $\tan(38.7^\circ) = \frac{8}{b}$

- b. Determine the value of b . Round to the nearest thousandth.

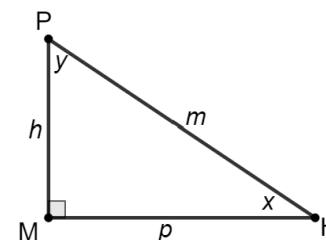
2. $\triangle SGK$ is shown.

- a. Use a trigonometric ratio for θ to write an equation that can be used to find the value of k .



- b. Which trigonometric ratio for θ cannot be used to write an equation that can be used to find the value of k ?

3. $\triangle PMH$ is shown. Work with your partner to determine which trigonometric ratio can be used to determine the unknown side or angle. Then, use the ratio to write an equation that can be used to determine the unknown. When writing the equation, use the variables h , p , m , x , and y .



Given	Unknown	Trigonometric Ratio	Equation
$m\angle PHM$ and length of $\overline{PM}$	$\overline{PH}$	☐ sin ☐ cos ☐ tan	
$m\angle PHM$ and length of $\overline{PM}$	$\overline{MH}$	☐ sin ☐ cos ☐ tan	
$m\angle HPM$ and length of $\overline{PM}$	$\overline{PH}$	☐ sin ☐ cos ☐ tan	
Lengths of $\overline{MH}$ and $\overline{PH}$	$m\angle HPM$	☐ sin ☐ cos ☐ tan	
Lengths of $\overline{MH}$ and $\overline{PM}$	$m\angle PHM$	☐ sin ☐ cos ☐ tan	



7.5.3 Guided Instruction: Using Inverse Trigonometric Functions to Find Angle Measure

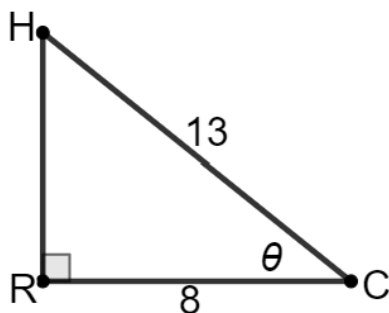


Inverse Trigonometric Functions

$\cos^{-1}\left(\frac{a}{c}\right)$ is arccosine. $\sin^{-1}\left(\frac{b}{c}\right)$ is arcsine.
 $\tan^{-1}\left(\frac{b}{a}\right)$ is arctangent. These functions can be used to determine the measure of an angle where a is the side adjacent the angle, b is the side opposite the angle, and c is the hypotenuse. The calculator should be in degree mode.

1. In $\triangle HRC$, $m\angle HRC = 90^\circ$, $RC = 8$, and $HC = 13$.

The equation $\cos(\theta) = \frac{8}{13}$ can be used to determine θ . To find θ using a calculator use the $\cos^{-1}\left(\frac{8}{13}\right)$.



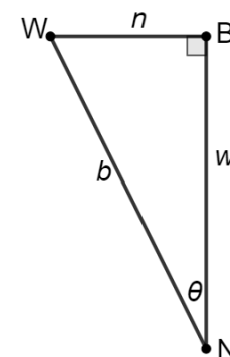
- What is $\cos^{-1}\left(\frac{8}{13}\right)$?
- Write an equation that can be used to find $m\angle CHR$.

- Find $m\angle CHR$ using an inverse trigonometric function on a calculator. Be sure the calculator is in degree mode.

2. $\triangle WLT$ is a right triangle with $m\angle WTL = 90^\circ$, $TL = 11$, and $WL = 23$. What is the $m\angle TWL$?

3. $\triangle WBN$ is shown, where $\angle B$ is a right angle.

- Determine the measure of $\angle BNW$ if $n = 36$ and $w = 77$.

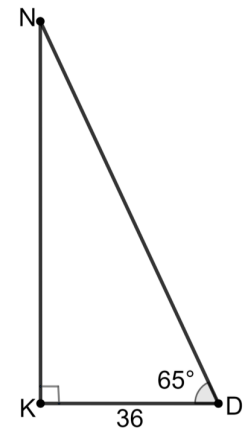


- b. Gabriella used $\sin^{-1}\left(\frac{77}{85}\right)$ to find the measure of $\angle BNW$. Determine if $\sin^{-1}\left(\frac{77}{85}\right)$ results in the same degree measure as your answer.
- c. Discuss with your partner whether or not Gabriella's method is valid. Write a justification of your decision.



7.5.4 Guided Instruction: Using Trigonometric Ratios to Find Side Measures

1. In $\triangle KDR$, $m\angle DKR = 31.2^\circ$, $m\angle KDR = 90^\circ$, and $KD = 17$. What is the length of $\overline{KR}$?
2. $\triangle MAU$ is a right triangle with $MA = 20.7$ and $MU = 9.1$. If $m\angle MUA = 90^\circ$, what is the $m\angle AMU$?
3. $\triangle NKD$ is shown, where $\angle K$ is a right angle, $KD = 36$, and $m\angle D = 65^\circ$
 - a. Find the length of $\overline{ND}$. Show your work.



- b. Amelia and Henry found the length of $\overline{NK}$ using two different methods. The first step of their work is shown. Complete their work.

Amelia	Henry
$36^2 + NK^2 = (85.183)^2$	$\tan(65^\circ) = \frac{NK}{36}$

- c. Describe some possible reasons that Amelia's and Henry's answers are not exactly the same value.
- d. Whose method gives a precise answer?



7.5.5 Guided Practice: Using Trigonometric Ratios

1. $\triangle ZME$ has $ME = 14.9$ and $m\angle ZEM = 41.5^\circ$.
- a. If $\angle ZME$ is a right angle, what is the length of $\overline{ZM}$?

- b. Sharif used the information to determine the length of $\overline{ZM}$. Her work is shown below.

$$\tan 41.5^\circ = \frac{14.9}{ZM}$$

$$ZM \cdot \tan 41.5^\circ = 14.9$$

$$ZM = \frac{14.9}{\tan 41.5^\circ}$$

$$ZM \approx 16.841$$

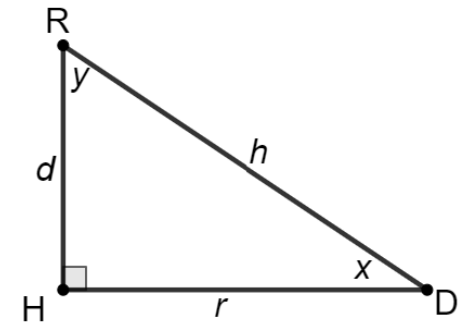
Find Sharif's error and correct it.

2. $\triangle KNP$ is a right triangle with $KP = 46$ and $KN = 27.3$. If $\angle PNK$ is a right angle, what is the $m\angle KPN$?



7.5.6 Wrap-Up: Ticket Out the Door

1. $\triangle RDH$ is shown, where $\angle RHD$ is a right angle.



- a. If $d = 2.8$ and $h = 4.6$, what is the measure of y , in degrees?
- b. If $x = 29^\circ$ and $h = 14.5$ what is the value of d ?

Unit 7, Lesson 6: Special Right Triangles



Benchmark

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.



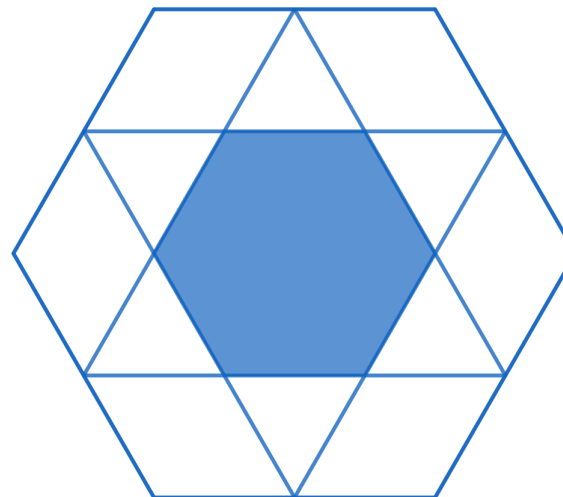
Learning Targets

- ☐ I can use side ratios of 45 – 45 – 90 triangles to solve problems.
- ☐ I can use side ratios of 30 – 60 – 90 triangles to solve problems.



7.6.1 Warm-Up: Critical Thinking

1. What fraction of the shape is shaded?

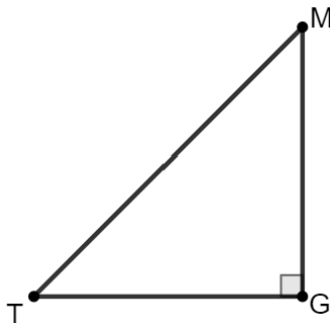


2. Describe your problem-solving method.



7.6.2 Exploration: Exploring Special Right Triangles

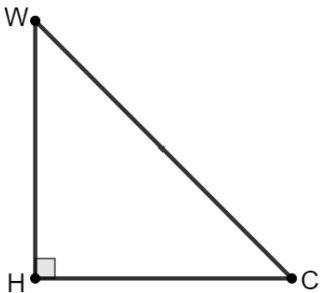
1. $\triangle MGT$ is a right triangle, where $m\angle MGT = 90^\circ$, $MG = 6$, and $GT = 6$.



- a. What type of right triangle is $\triangle MGT$?
- b. Determine the measurements. Leave the measurement of $\overline{MT}$ in radical form.

MT	$m\angle GMT$	$m\angle GTM$

2. $\triangle WCH$ is a right triangle, where $m\angle WHC = 90^\circ$, $m\angle WCH = 45^\circ$, and $WC = 4\sqrt{2}$.



- a. What type of right triangle is $\triangle WCH$?
- b. Determine the measurements.

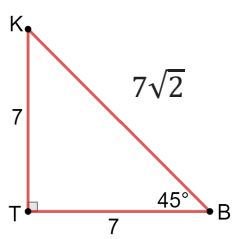
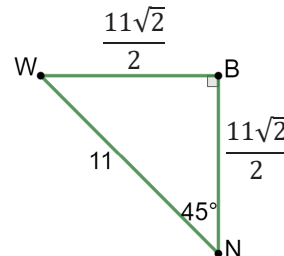
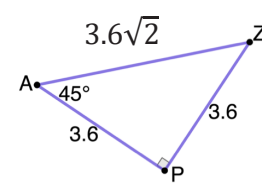
HC	WH	$m\angle CWH$

3. What do you notice about the angles of $\triangle WCH$ and $\triangle MGT$?

4. What do you notice about the legs of $\triangle WCH$ and $\triangle MGT$?

$\triangle WHC$ and $\triangle MGT$ are $45 - 45 - 90$ triangles.

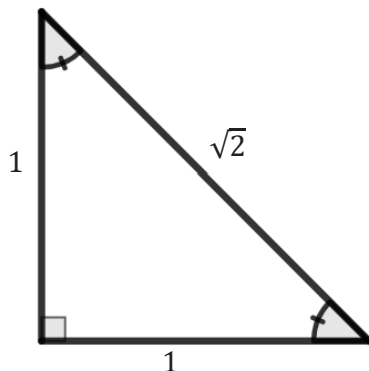
5. The table shows three $45 - 45 - 90$ triangles.

$\triangle KBT$	$\triangle WBN$	$\triangle AZP$
		

- Discuss with your partner the relationship between the legs of a $45 - 45 - 90$ triangle and its hypotenuse. Write a conjecture.
- Ask two other pairs of partners for the conjecture they wrote. Listen to their conjecture and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was accurate.

Conjecture	Initials

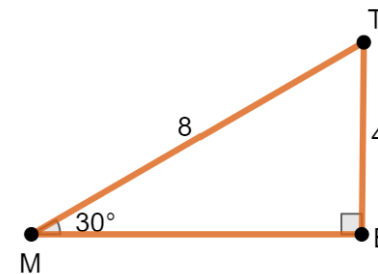
6. An isosceles right triangle is shown.



Describe how the triangle shows the relationships in a 45 – 45 – 90 triangle.

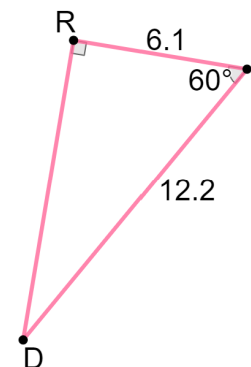
7. $\triangle MBT$ is a right triangle, where $m\angle MBT = 90^\circ$, $m\angle TMB = 30^\circ$, $TB = 4$, and $MT = 8$.

Find the length of $\overline{MB}$ using the Pythagorean Theorem. Write the answer in simplified radical form.



8. $\triangle RFD$ is a right triangle, where $m\angle DRF = 90^\circ$, $m\angle RFD = 60^\circ$, $RF = 6.1$, and $FD = 12.2$.

Discuss with your partner what features $\triangle RFD$ has in common with $\triangle MBT$.



9. $\triangle MBT$ and $\triangle RFD$ are 30 – 60 – 90 triangles.

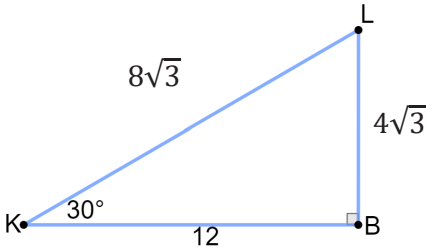
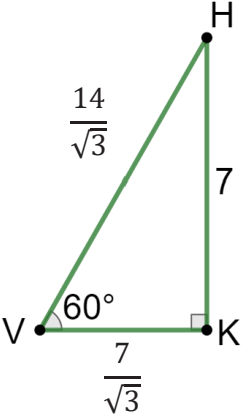
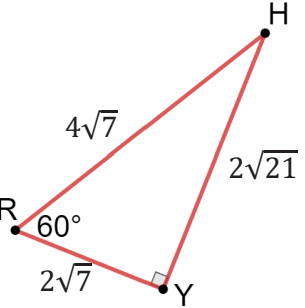
The table gives the exact measurements for each of the sides of $\triangle MBT$ and $\triangle RFD$.

Triangle	Short Leg	Long Leg	Hypotenuse
$\triangle MBT$	4	$4\sqrt{3}$	8
$\triangle RFD$	6.1	$6.1\sqrt{3}$	12.2

Complete the table by describing the relationship between the side lengths of a 30 – 60 – 90 triangle.

Sides	Relationship
short leg and hypotenuse	
short leg and long leg	

10. The table below includes three 30 – 60 – 90 triangles.

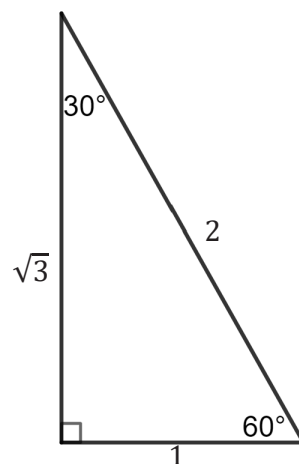
$\triangle KBL$	
$\triangle VHK$	
$\triangle HRY$	



Discuss with your partner the relationships between the sides of a 30 – 60 – 90 triangle in the table. Complete the table by showing how the short side of each of the triangles can be used to determine the long side and the hypotenuse. Write the final length as an exact value.

Triangle	Short Side	Work to Find the Long Side	Work to Find the Hypotenuse
ΔKBL	$4\sqrt{3}$		
ΔVHK	$\frac{7}{\sqrt{3}}$		
ΔHRY	$2\sqrt{7}$		

11. Describe how the triangle shows the relationships in a 30 – 60 – 90 triangle.

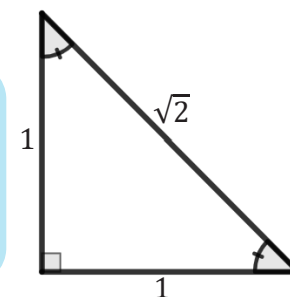


7.6.3 Guided Instruction: 45-45-90 and 30-60-90 Triangles



The side lengths of a 45 – 45 – 90 triangle can be found using the equation

$$\text{leg} \cdot \sqrt{2} = \text{hypotenuse}.$$

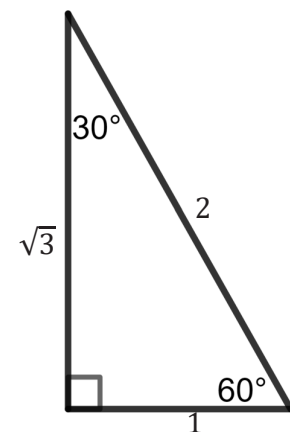


1. ΔZYN has $m\angle YZN = m\angle YNZ = 45^\circ$, $m\angle ZYN = 90^\circ$, and $ZN = 12$. What are the lengths of the legs of the triangle? Leave your answer as an exact answer.



The lengths of the short leg and the long leg of a

30 – 60 – 90 triangle can be found using the equation $\text{short leg} \cdot \sqrt{3} = \text{long leg}$.



The lengths of the short leg and the hypotenuse of a 30 – 60 – 90 triangle can be found using the equation $\text{short leg} \cdot 2 = \text{hypotenuse}$.

2. $\triangle BHK$ is a $30 - 60 - 90$ triangle with $m\angle KBH = 30^\circ$, $m\angle BKH = 90^\circ$, and $BK = 5\sqrt{3}$. What are the measures of the other sides of the triangle?



7.6.4 Guided Practice: Solving Special Right Triangles

1. $\triangle LPM$ is a $45 - 45 - 90$ triangle with $m\angle LMP = m\angle MLP = 45^\circ$, $m\angle LPM = 90^\circ$, and $LM = 7$. What are the lengths of $\overline{PM}$ and $\overline{LP}$?
2. $\triangle TNW$ is a right triangle with $m\angle NTW = 60^\circ$, $m\angle WNT = 90^\circ$, and $TW = 18$. What are the measures of the other sides of the triangle?



7.6.5 Wrap-Up: Ticket Out the Door

1. $\triangle FHX$ has angle measures $m\angle HXF = 90^\circ$ and $m\angle HFX = 30^\circ$, and side length $HX = 6\sqrt{3}$. What are the exact measures of $\overline{FX}$ and $\overline{FH}$?

Unit 7, Lesson 7: The Unit Circle and Its Coordinates



Benchmarks

MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles.

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.



Learning Targets

- ☐ I can explore relationships between the coordinates of vertices when the triangles are in different quadrants.
- ☐ I can use side ratios of 45 – 45 – 90 triangles to solve problems.
- ☐ I can use side ratios of 30 – 60 – 90 triangles to solve problems.



7.7.1 Warm-Up: Between Two Numbers



1. A Florida Box Turtle's average speed is about 0.17 miles per hour.

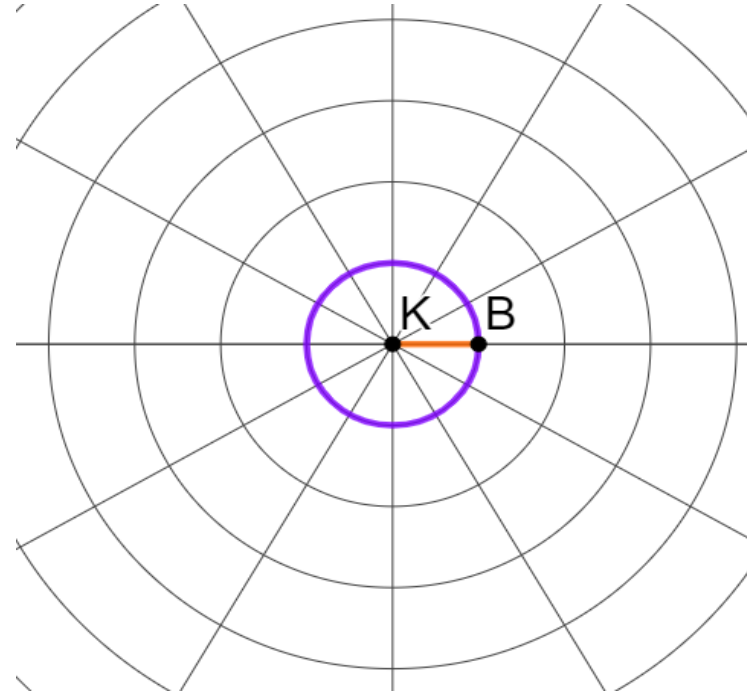
The North American X-15's top speed is about 163 miles per hour.

How much time would it take for each to travel across Florida, a distance of about 500 miles?



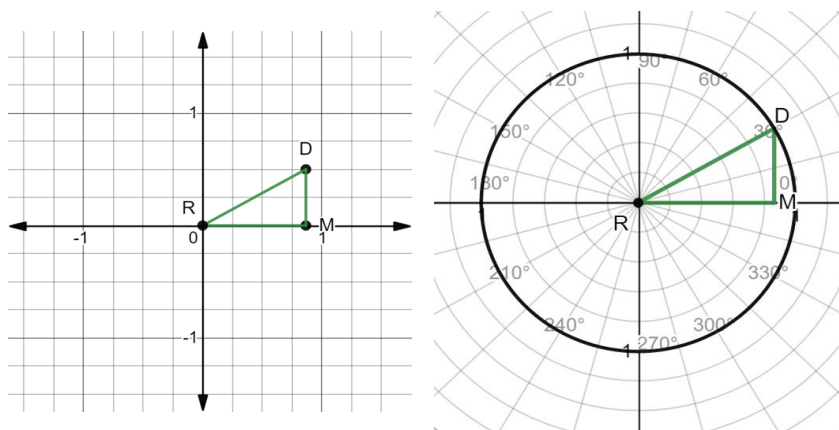
7.7.2 Exploration: The Unit Circle

1. Circle K is shown on polar graph paper, where the length of $\overline{KB}$ is 1 unit.



- a. Draw a circle that has a radius of 2 units.
- b. Draw a circle that has a radius of 3 units.

2. $\triangle RDM$ is a right triangle formed by drawing a 30° angle at the origin. $\triangle RDM$ is shown on both a coordinate grid and on a polar graph, where $m\angle MRD = 30^\circ$, $m\angle RDM = 60^\circ$, $m\angle DMR = 90^\circ$, $RD = 1$, $DM = \frac{1}{2}$, and $RM = \frac{\sqrt{3}}{2}$.



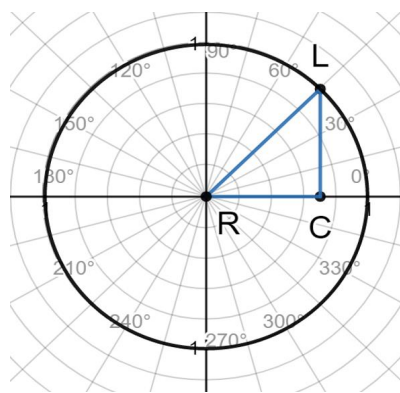
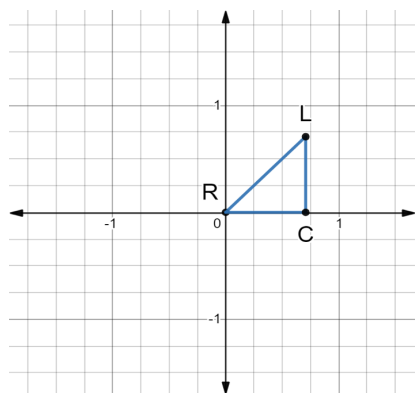
- a. Discuss with your partner any similarities and differences between the graphs. Summarize your discussion.
- b. Work with your partner to determine the coordinates of point D .

- c. Work with your partner to complete the table. Leave answers in exact form.

Angle	Cosine	Sine	Tangent
30° ($m\angle R$)			

- d. Discuss with your partner how the coordinates of point D can be found using the values from the table. Make a conjecture about the relationship between the values of cosine and sine, and the coordinate pair formed by the angle.

3. $\triangle RLC$ is a right triangle formed by drawing a 45° angle at the origin. $\triangle RLC$ is shown on both a coordinate grid and on a polar graph, where $m\angle CRL = 45^\circ$, $m\angle RLC = 45^\circ$, $m\angle RCL = 90^\circ$, $RL = 1$, $RC = \frac{\sqrt{2}}{2}$, and $LC = \frac{\sqrt{2}}{2}$.



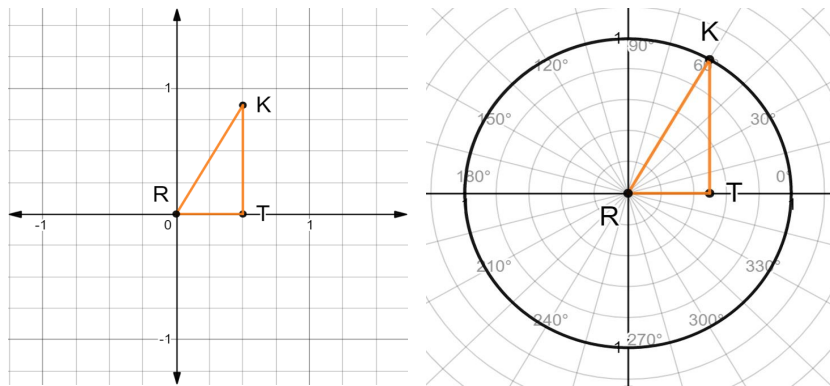
- a. Determine the coordinates of point L .

- b. Work with your partner to complete the table.
Leave answers in exact form.

Angle	Sine	Cosine	Tangent
45° ($m\angle R$)			

- c. Discuss with your partner whether or not the conjecture that was made about the coordinates of point D could be applied to determine the coordinates of point L . If necessary, revise your conjecture.

4. $\triangle RKT$ is a right triangle formed by drawing a 60° angle at the origin. $\triangle RKT$ is shown on both a coordinate grid and on a polar graph, where $m\angle TRK = 60^\circ$, $m\angle RKT = 30^\circ$, $m\angle RTK = 90^\circ$, $RK = 1$, $RT = \frac{1}{2}$, and $TK = \frac{\sqrt{3}}{2}$.



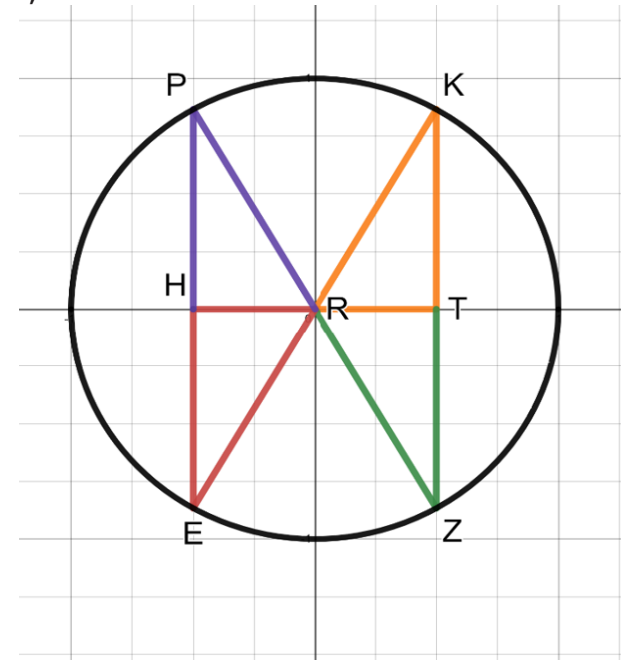
a. Determine the coordinates of point K .

- b. Work with your partner to complete the table. Leave answers in exact form.

Angle	Sine	Cosine	Tangent
60° ($m\angle R$)			

- c. Discuss with your partner how the coordinates will change if the triangle is in a different quadrant. Make a prediction.

5. Four triangles were formed by drawing angles of 60° from the origin. $m\angle TRK = 60^\circ$, $m\angle PRH = 60^\circ$, $m\angle HRE = 60^\circ$, and $m\angle TRZ = 60^\circ$. $RK = 1$, $RP = 1$, $RE = 1$, and $RZ = 1$.

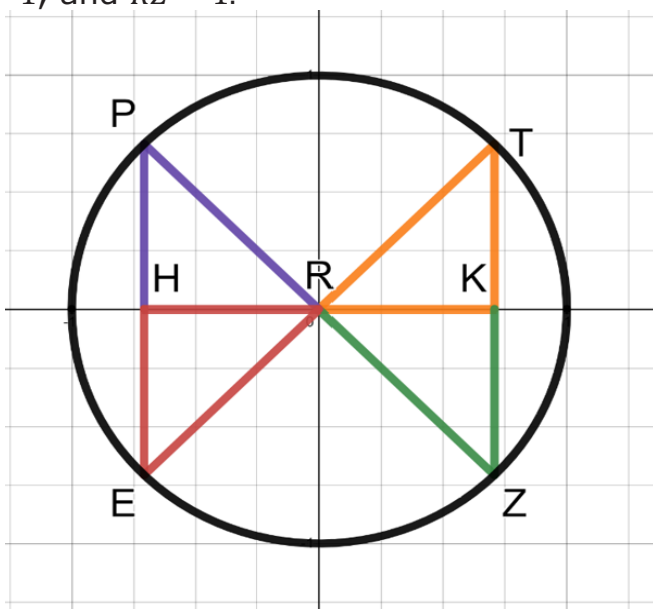


a. Complete the table.

Point	Coordinates
K	$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
P	
E	
Z	

- b. Points K , P , E , and Z lie on circle R . What is the radius of circle R ?
- c. What type of special triangles are ΔRKT , ΔRPH , ΔREH , and ΔRZT ?

6. Four triangles were formed by drawing angles of 45° from the origin. $m\angle TRK = 45^\circ$, $m\angle PRH = 45^\circ$, $m\angle HRE = 45^\circ$, and $m\angle KRZ = 45^\circ$. $RT = 1$, $RP = 1$, $RE = 1$, and $RZ = 1$.

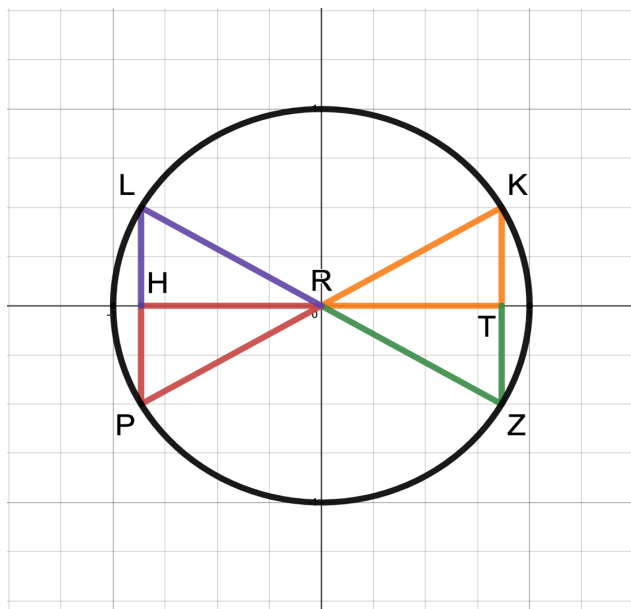


- a. Complete the table.

Point	Coordinates
T	$\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$
P	
E	
Z	

- b. Points T , P , E , and Z lie on circle R . What is the radius of circle R ?
- c. What type of special triangles are ΔRTK , ΔRPH , ΔREH , and ΔRKZ ?

7. Four triangles were formed by drawing angles of 30° from the origin. $m\angle TRK = 30^\circ$, $m\angle PRH = 30^\circ$, $m\angle HRL = 30^\circ$, and $m\angle TRZ = 30^\circ$. $RK = 1$, $RP = 1$, $RL = 1$, and $RZ = 1$.



- a. Complete the table.

Point	Coordinates
K	$\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$
L	
P	
Z	

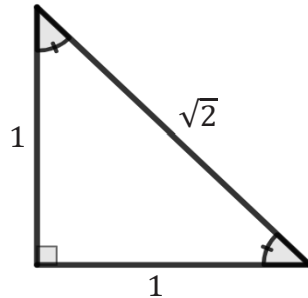
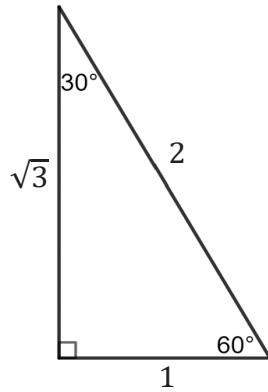
- b. Points K , P , L , and Z lie on circle R . What is the radius of circle R ?
- c. What type of special triangles are ΔRKT , ΔRPH , ΔRHL , and ΔRZT ?

8. For each type of special triangle, what similarities do the coordinates for each vertex located on the circle share with one another?
9. For each type of special triangle, what differences do the coordinates for each vertex located on the circle share with one another?



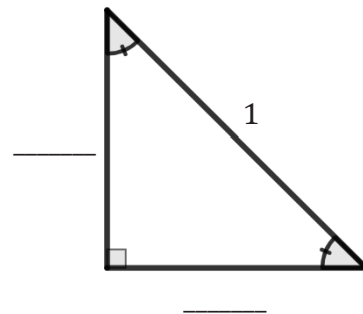
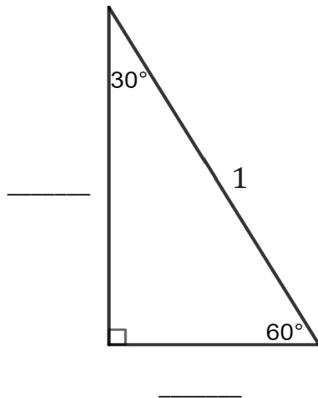
7.7.3 Bring It Together: The Unit Circle

In the Exploration, all of the triangles were one of the two special triangles, $30 - 60 - 90$ or $45 - 45 - 90$. The relationships of each of the side lengths for these triangles are shown.

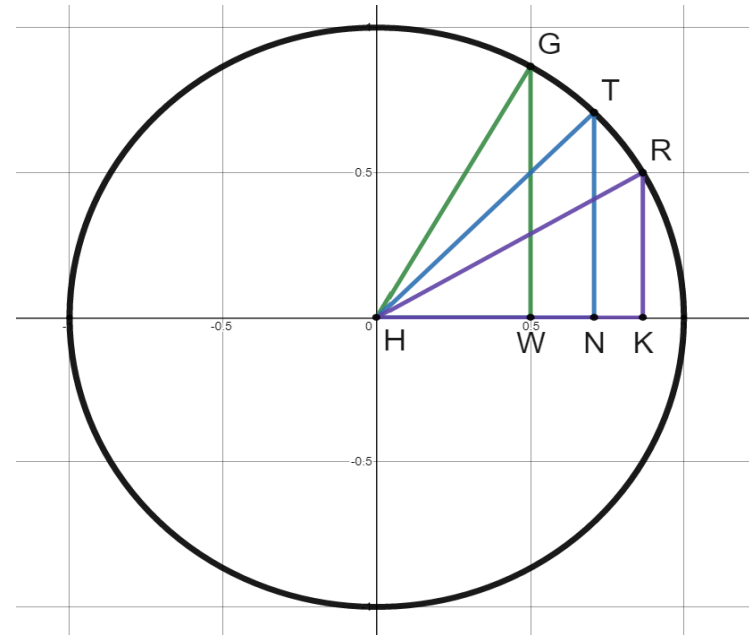


In the circle, the hypotenuse of the triangle was the radius of the circle, which was 1. When a circle has a radius of 1 and is centered at the origin, it is called the unit circle. In the diagrams of the two special triangles the hypotenuse is not 1.

- Fill in the blanks to show the relationships of the special triangles when the hypotenuse is 1.



- A unit circle with $\triangle HKR$, $\triangle HGW$, and $\triangle HNT$ is shown, where $m\angle KHR = 30^\circ$, $m\angle NHT = 45^\circ$, $m\angle WHG = 60^\circ$, $m\angle GWH = 90^\circ$, $m\angle HNT = 90^\circ$, and $m\angle HKR = 90^\circ$.



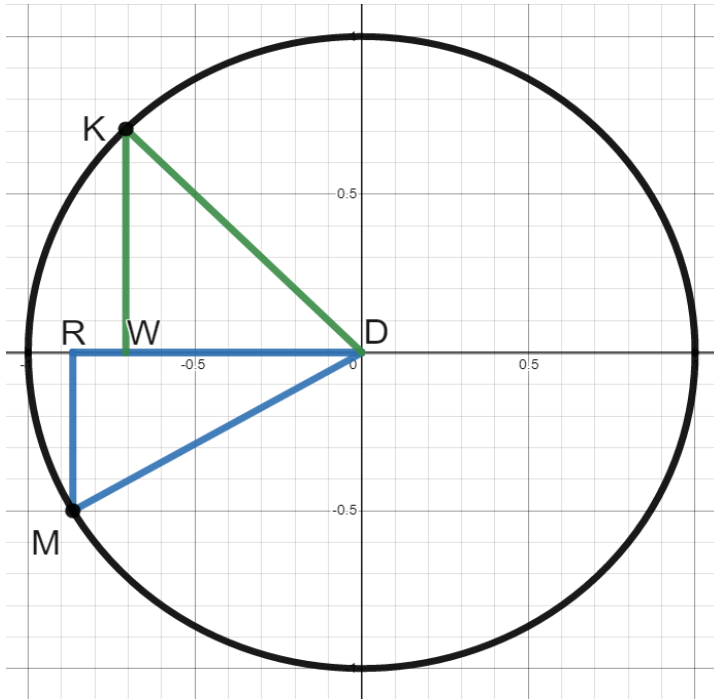
Complete the table.

Point	Coordinates
G	
R	
T	



7.7.4 Try It: Finding Coordinates in the Unit Circle

1. A unit circle is shown where $\triangle KDW$ and $\triangle MRD$ are right triangles, $m\angle KDW = 45^\circ$, and $m\angle RDM = 30^\circ$.



Complete the table.

Point	Coordinates
K	
M	



7.7.5 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the Exploration that I knew were ...	Something I wondered during the Exploration was ...	Something I learned during the Exploration was ...

Unit 7, Lesson 8: Using Trigonometric Ratios and the Pythagorean Theorem



Benchmark

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.



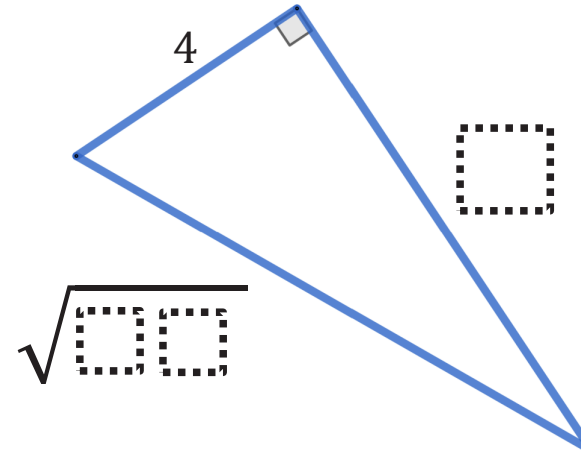
Learning Targets

- ☐ I can use trigonometric ratios to solve mathematical problems.
- ☐ I can use the Pythagorean Theorem to solve mathematical problems.



7.8.1 Warm-Up: Critical Thinking

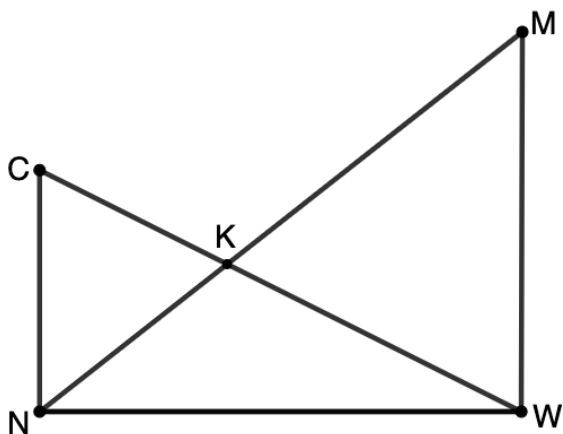
- Use the digits 0-9 one time each to find the missing side lengths of the right triangle. The triangle is not drawn to scale.



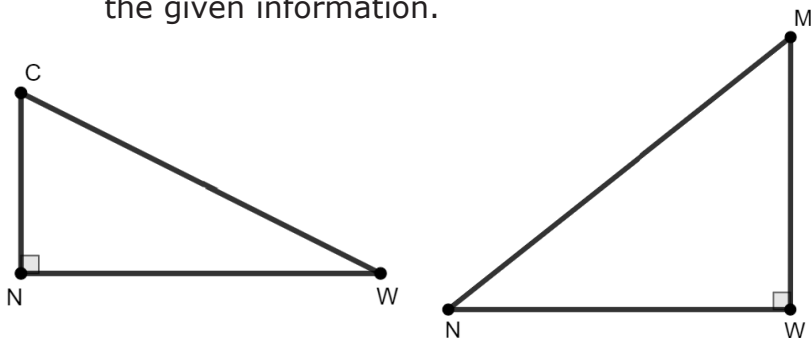


7.8.2 Collaboration: Overlapping Triangles

1. $\triangle MWN$ and $\triangle CNW$ are right triangles that share side $\overline{NW}$ and K is the intersection point of sides $\overline{CW}$ and $\overline{MN}$. In the triangles, the $m\angle MWN = m\angle CNW = 90^\circ$, $MW = 4.4$, $NW = 6$, and $m\angle NCK = 63^\circ$. $\triangle CNK$ is an isosceles triangle with base $\overline{CK}$ and $\triangle MWK$ is an isosceles triangle with base $\overline{WK}$.



- a. Mark the figure with all information given in the problem.
- b. Jasmine redrew the diagram as two separate triangles, as shown. Mark Jasmine's diagram using the given information.



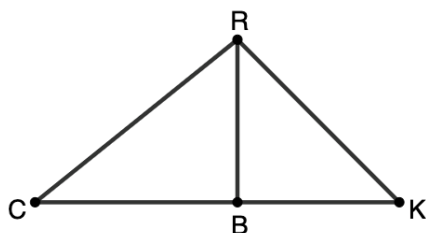
- c. Find the measures of the segments and angles listed in the table. Round to the nearest thousandth.

Segment	Length	Angle	Measure
$\overline{MN}$		$\angle MNW$	
$\overline{CN}$		$\angle WMN$	
$\overline{CW}$		$\angle CWN$	
$\overline{KN}$		$\angle WCN$	
$\overline{MK}$		$\angle CKN$	



7.8.3 Guided Instruction: Solving Problems With Right Triangles

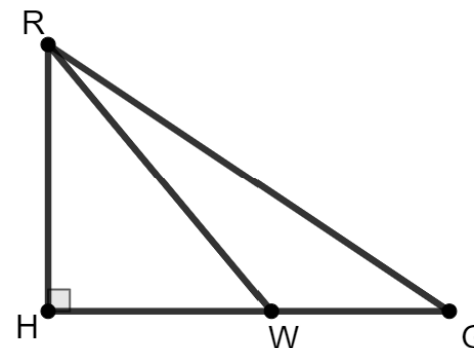
1. $\triangle RBK$ and $\triangle RBC$ are right triangles that share side $\overline{RB}$. $\angle RBK$ and $\angle RBC$ are both right angles, $m\angle BCR = 38.7^\circ$, $m\angle RKC = 45^\circ$, and $RB = 8$.
- a. Use the given information to complete the table. Whenever possible, give lengths as exact values. If necessary, round to the nearest hundredth.



Segment	Length
$\overline{RC}$	
$\overline{CB}$	
$\overline{RK}$	
$\overline{KB}$	
$\overline{CK}$	

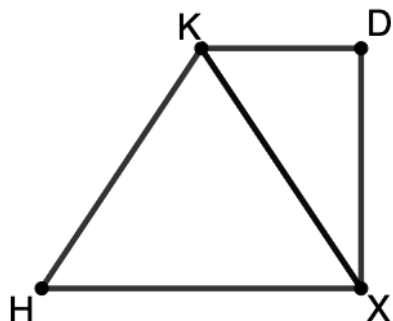
- b. What is the area of $\triangle CRK$?

2. $\triangle RGH$ is shown, where $m\angle RHG = 90^\circ$, $m\angle RWH = 50.2^\circ$, $m\angle WRG = 16.5^\circ$, $m\angle RGH = 33.7^\circ$, and $RH = 12$.



What is the approximate perimeter of $\triangle RWG$? Round your answer to the nearest hundredth.

3. $\triangle KDX$ and $\triangle HKX$ share side $\overline{KX}$. $\triangle HKX$ is an isosceles triangle with base $\overline{HX}$ and $\triangle KDX$ is a right triangle with $m\angle KDX = 90^\circ$.

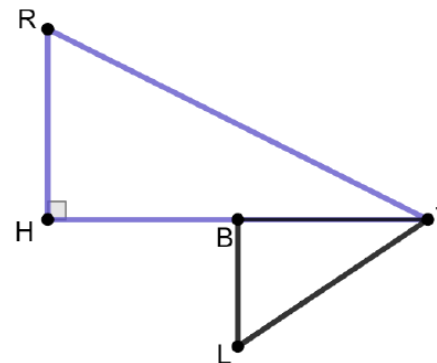


If $HX = 13$, $DX = 9$, and $m\angle KXD = 34^\circ$, what is the perimeter of quadrilateral $KDXH$? Round your answer to the nearest hundredth.



7.8.4 Guided Practice: Right Triangle Trigonometry and the Pythagorean Theorem

1. $\triangle RHT$ and $\triangle TBL$ are right triangles, where B is the midpoint of $\overline{HT}$, $m\angle RHT = m\angle TBL = 90^\circ$, $m\angle RTH = 30^\circ$, $BL = 3$, and $LT = 7$.

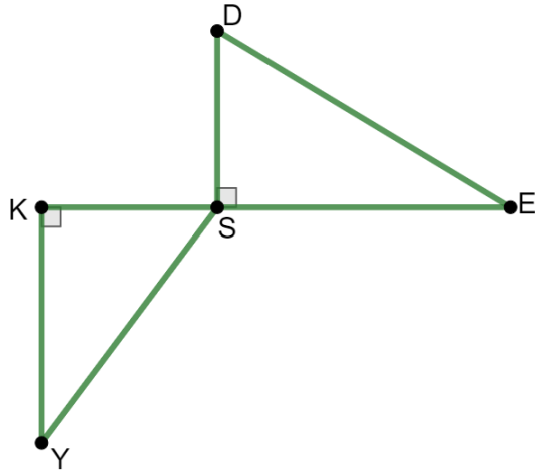


- What is the length of $\overline{HT}$?
- What is the area of $\triangle RHT$?



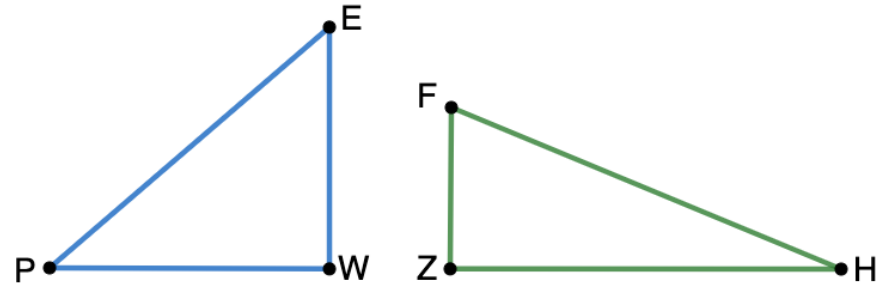
7.8.5 Your Turn!

1. In the diagram, $\triangle DSE$ and $\triangle YKS$ are right triangles that share the vertex S . The $m\angle DSE = m\angle YKS = 90^\circ$, $m\angle DES = 31^\circ$, $m\angle KSY = 53^\circ$, $KE = 37$, and $DS = 10$.



Determine the length of $\overline{KY}$.

2. $\triangle PWE$ and $\triangle FZH$ are shown. In $\triangle PWE$, $m\angle WEP = 49.3^\circ$, $PE = 30.6$, and $\angle PWE$ is a right angle.



If $\angle FZH$ is a right angle, $\overline{FZ}$ is two-thirds the length of $\overline{EW}$, and $\overline{ZH}$ is 1.2 times longer than $\overline{PW}$, what is the length of $\overline{FH}$?



7.8.6 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Unit 7, Lesson 9: Using Trigonometry in the Real World



Benchmark

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.



Learning Target

- ☐ I can solve real-world problems using trigonometric ratios and the Pythagorean Theorem.



7.9.1 Warm-Up: Notice and Wonder

1. The picture shows the stairs in the Ponce de Leon Inlet Lighthouse.



- a. What do you notice?
- b. What do you wonder?
- c. The lighthouse has 203 steps. Do you think this picture includes all of the steps? Explain your answer.



7.9.2 Collaboration: Angle of Depression and Angle of Elevation

Real-world trigonometric ratio problems involve angles that are made from a horizontal line of sight.

1. Using a ruler, draw the horizontal line of sight for the man.

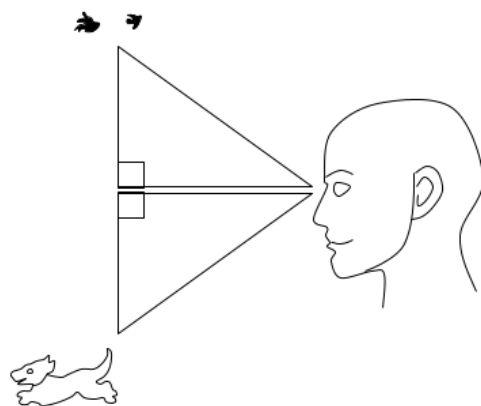


2. The horizontal line of sight is always

- ☐ perpendicular
 - ☐ parallel

 to the ground.
3. The angles created from the horizontal line of sight are called angles of elevation and angles of depression. In the image, Hector is looking at birds in the sky and a dog on the ground.

Label the angle of elevation with an E , label the angle of depression with a D .



4. Based on the diagram, the name of the angle, work with a partner to write a definition for an angle of elevation.

5. Work with a partner to write a definition for an angle of depression.

6. Compare your definitions with that of another pair. Discuss any differences. Make changes as necessary.



7.9.3 Guided Instruction: Angles of Elevation and Depression

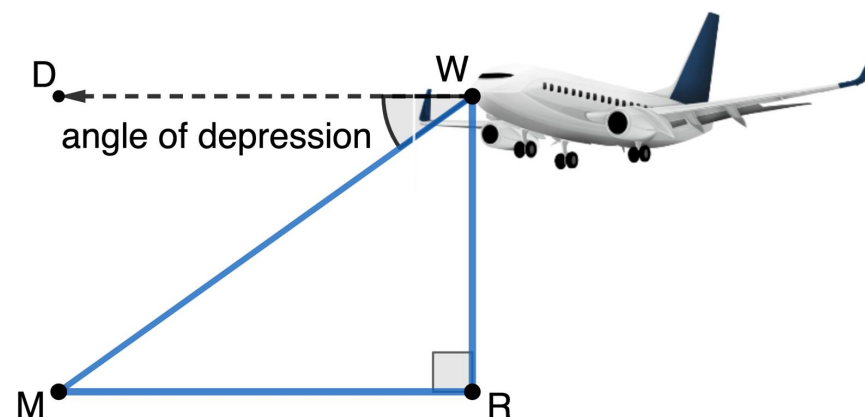


The **angle of elevation** is the angle created above the horizontal when looking up at an object.



The **angle of depression** is the angle created below the horizontal when looking down at an object.

1. The angle of depression, $m\angle DWM$, from the airplane to the runway, point M , is 37° .

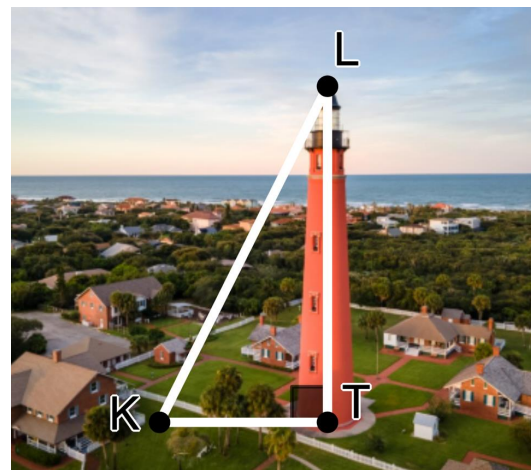


- a. Explain why $\angle DWM$ is congruent to $\angle RMW$.

- b. If the airplane is flying at an altitude of 2500 feet, WR , and is descending to the airport, M , what is the distance, MW , the plane has to travel?

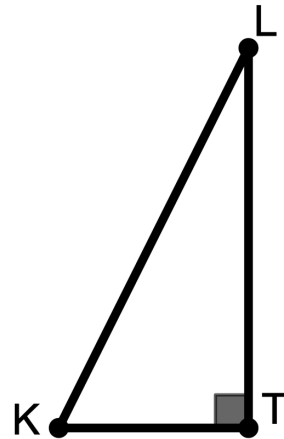
- c. How far is the runway from a point on the ground directly below the airplane, MR ?

2. Kamala is visiting the Ponce de Leon Inlet Lighthouse. Kamala is standing at point K and is looking at the top of the lighthouse, point L . $\overline{KT}$ is from Kamala's eye level, which is at 1.75 meters (m).



- a. When Kamala looks at the top of the lighthouse, is the angle created an angle of elevation or depression?

- b. The height of the Ponce de Leon Inlet Lighthouse is 53 m. Kamala is standing 15.25 m from the lighthouse. Label $\triangle KTL$ appropriately with this information.

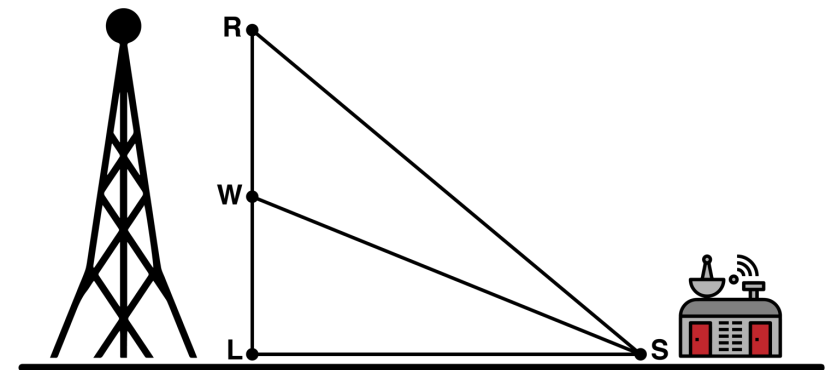


- c. What is the measure of the angle created when Kamala looks at the top of the lighthouse?



7.9.4 Guided Practice: Trigonometry in the Real World

1. A cell phone tower was built in two sections, with a control shed built 60 feet from the base of the tower on level ground. From the base of the control shed, the angle of elevation, $\angle LSW$, to the top of the first section of tower, $\overline{LW}$, is 22° . The angle of elevation to the top of the second section, $\angle LSR$, is 46° .



- Label the diagram with the information from the problem.
- Determine the length of $\overline{RW}$, the second section of tower. Round your answer to the nearest thousandth of a foot.

2. A hot air balloon is flying 38.5 ft above level ground. The Orlando City Soccer field is 200 ft from the point on the ground directly below the balloon.

a. Draw and label a diagram using the information from the problem.

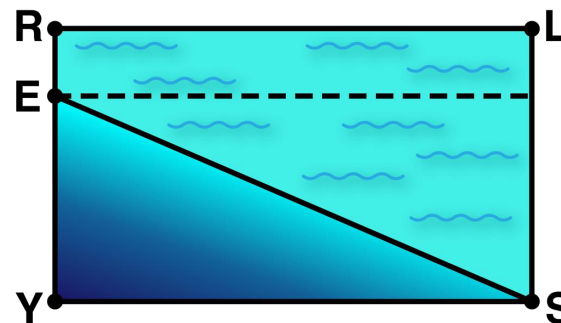
b. What is the angle of elevation from the soccer field to the hot air balloon?

c. The hot air balloon ascends further into the air. If the new angle of elevation from the soccer field to the balloon is 17.3° . How many feet did the hot air balloon rise?



7.9.5 Your Turn!

1. A swimming pool is 48 feet long and 4 feet deep in the shallow end. There is a steady decline of 14° toward the deep end of the pool. Rectangle $RLSY$ and $\triangle SYE$ are shown to model the transition from the shallow end to the deep end. Label the diagram appropriately.



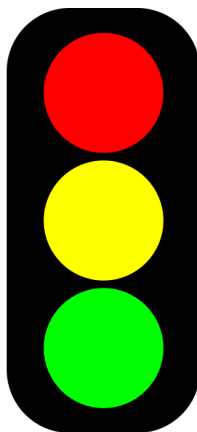
a. How deep is the pool at the deep end?

b. $\overline{ES}$ represents the slant in the pool from the shallow end to the deep end. What is its length?



7.9.6 Wrap-Up: Reflection

1. Circle the color on the stoplight that best represents your understanding of solving real-world problems using an angle of depression or an angle of elevation.



Red: I do not understand.

Yellow: I need some more help.

Green: I got this; I understand!

2. Complete the following statements if you selected red or yellow.
 - a. I am still unsure about ...
 - b. To fill in this gap I intend to ...



Unit 7, Lesson H1: Law of Sines



Benchmark

MA.912.T.1.3 Apply the Law of Sines and the Law of Cosines to solve mathematical and real-world problems involving triangles.



Learning Target

- ☐ I can solve mathematical and real-world problems involving triangles using the Law of Sines.



7.H1.1 Warm-Up: Notice and Wonder

1. A theodolite is an instrument used for measuring angles between designated visible points on the horizontal and vertical planes. Theodolites are often used for land surveying.

Late 16 th century Theodolite	Modern Day Theodolite
	

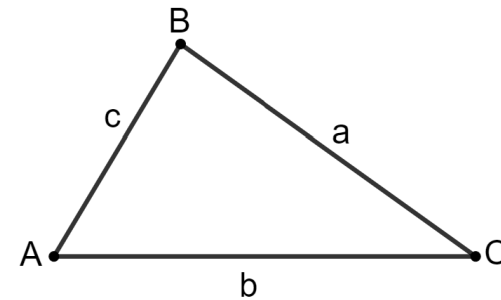
- a. What do you notice?
- b. What do you wonder?



7.H1.2 Collaboration: Deriving the Formula for Law of Sines

Right triangle trigonometry can be used to solve problems involving right triangles, but not all triangles are right triangles. If a triangle is not a right triangle, the triangle's altitude can be used to create a right triangle so that right triangle trigonometry can be used.

1. $\triangle ABC$ is shown, where $AB = c$, $AC = b$, and $BC = a$.



- a. Use $\triangle ABC$ to do the following.
 - Sketch an altitude from vertex B .
 - Label the altitude k .

The altitude creates two right triangles inside $\triangle ABC$. Notice that $\angle A$ is contained in one of the right triangles, and $\angle C$ is contained in the other.

- b. Complete each equation using right triangle trigonometry.

$$\sin(A) =$$

$$\sin(C) =$$

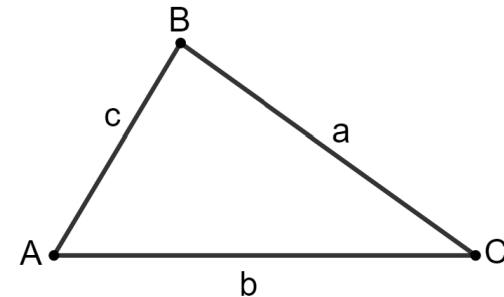
c. Discuss with your partner why both equations use k . Summarize your discussion.

d. Solve each equation for k .

e. Set the equations equal to each other to form a new equation.

f. Rewrite the new equation by regrouping a with $\sin(A)$ and c with $\sin(C)$.

2. $\triangle ABC$ is shown where $AB = c$, $AC = b$, and $BC = a$.



a. Use $\triangle ABC$ to do the following.

- Sketch an altitude from vertex C .
- Label the altitude k .

The altitude creates two right triangles inside $\triangle ABC$. Notice that $\angle A$ is contained in one of the right triangles, and $\angle B$ is contained in the other.

b. Complete each equation using right triangle trigonometry.

$$\sin(A) =$$

$$\sin(B) =$$

c. Solve each equation for k .

- d. Set the equations equal to each other to form a new equation.
- e. Re-write the new equation by regrouping a with $\sin(A)$ and b with $\sin(B)$.



The **Law of Sines** is $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

The Law of Sines may also be written as

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$



7.H1.3 Guided Instruction: Law of Sines

1. Some of the measurements of $\triangle DLC$ are $DL = 17$, $m\angle DLC = 40^\circ$, and $m\angle CDL = 75^\circ$. Determine the measurement of each part of $\triangle DLC$. Round to the nearest tenth.

Part	Measurement
$\overline{LC}$	
$\overline{DC}$	
$\angle DCL$	

2. In $\triangle RFM$, $RM = 66$, $FM = 30$, and $m\angle RFM = 26^\circ$. Determine the measurement of each part of $\triangle RFM$. Round to the nearest tenth.

Part	Measurement
$\overline{FR}$	
$\angle FRM$	
$\angle FMR$	



7.H1.4 Guided Practice: Law of Sines

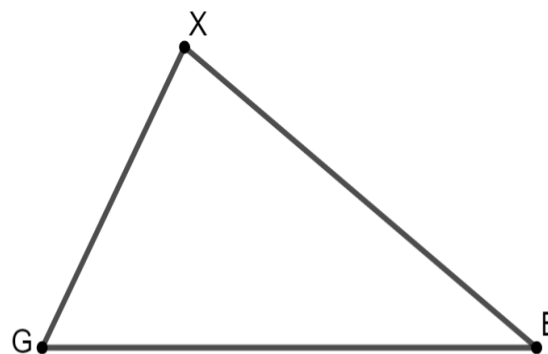
1. $\triangle LVY$ has side measurements of $VY = 8.5$ and $LY = 16.5$. The measure of $\angle LVY$ is 84° . Find all missing sides and angles. Round to the nearest tenth.

2. The three known measurements of $\triangle DNH$ are $DN = 26$, $m\angle NDH = 80^\circ$, and $m\angle DNH = 75^\circ$. Find all missing sides and angles. Round to the nearest tenth.



7.H1.5 Your Turn!

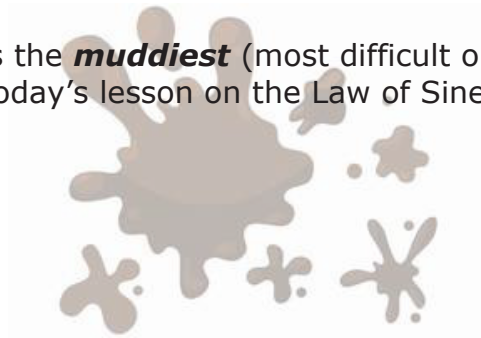
1. In $\triangle LSK$, $LS = 7$, $m\angle KLS = 31^\circ$, and $m\angle KSL = 107^\circ$. Determine the length of $\overline{KS}$. Round to the nearest tenth.
2. Cell phone towers represented by points G and B are 25 miles apart. A new tower, X , is being built so that $m\angle G = 54^\circ$ and $m\angle B = 26^\circ$. How far from tower G will tower X be built? Round to the nearest tenth.





7.H1.6 Wrap-Up: Reflection

1. What was the ***muddiest*** (most difficult or confusing) point in today's lesson on the Law of Sines?



Unit 7, Lesson H2: Law of Cosines



Benchmark

MA.912.T.1.3 Apply the Law of Sines and the Law of Cosines to solve mathematical and real-world problems involving triangles.



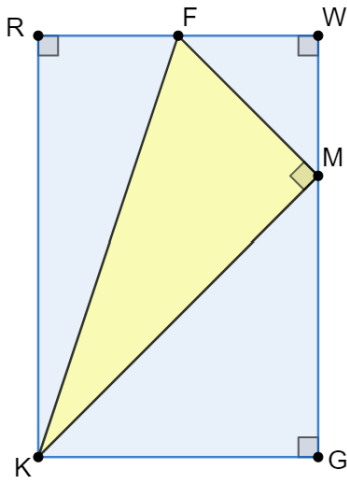
Learning Target

- ☐ I can solve mathematical and real-world problems that involve triangles using the Law of Cosines.



7.H2.1 Warm-Up: Reasoning with Triangles

1. A diagram of a triangle inscribed in a rectangle is shown.

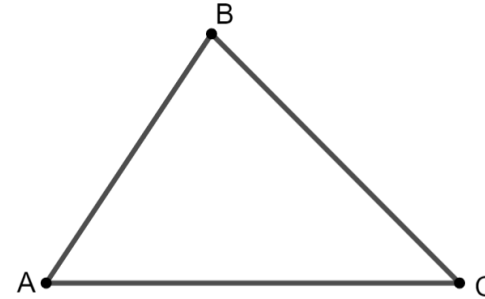


- Liu Yang told her partner that all of the right triangles in the diagram are similar. Determine if the diagram contains enough information to verify Liu Yang's statement. Explain your reasoning.
- In the diagram, $m\angle RFK = 72^\circ$ and $m\angle MKF = 63^\circ$. Use these measures to evaluate your reasoning. If necessary, make adjustments.



7.H2.2 Collaboration: Deriving the Law of Cosines

1. A triangle is shown below.



- Use $\triangle ABC$ to complete the following.
 - Sketch an altitude from vertex C .
 - Label the altitude h , and label the point on $\overline{AB}$ as D .
 - Label each line segment. $AC = b$, $AD = x$, $CB = a$, $DB = c - x$, $AB = c$

The altitude creates two right triangles inside $\triangle ABC$. Notice that $\angle A$ is contained in one of the right triangles, and $\angle B$ is contained in the other.

- Use the labels on the two right triangles inside $\triangle ACB$ to complete the table.

$\triangle ADC$	$\triangle CDB$
$\underline{\hspace{1cm}}^2 + \underline{\hspace{1cm}}^2 = \underline{\hspace{1cm}}^2$	$\underline{\hspace{1cm}}^2 + \underline{\hspace{1cm}}^2 = \underline{\hspace{1cm}}^2$

c. What do you notice about both equations?

d. Rearrange the Pythagorean Theorem for $\triangle ADC$ in terms of h^2 .

e. Substitute the new equation in place of h^2 in $\triangle CDB$.

f. Use cosine to write the trigonometric ratio for $\angle A$.

g. Solve the trigonometric ratio for $\angle A$ for x .

h. Substitute $b \cos A$ in place of x in the equation from part e.

i. Expand and combine like terms.



7.H2.3 Guided Instruction: Law of Cosines



The **Law of Cosines** can be written as one of the following:

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

1. In $\triangle DRP$, $DR = 10$, $DP = 4.7$, and $m\angle RDP = 60^\circ$.
 - a. Draw and label a triangle with the given measurements.
 - b. Use the law of cosines to find the measure of $\overline{RP}$.
 - c. Use the law of cosines to find $m\angle RPD$.

2. The measurements of the sides of $\triangle RLP$ are $RL = 4.5$, $PL = 7.2$, and $RP = 6.7$. Determine the measures of the angles of $\triangle RLP$.

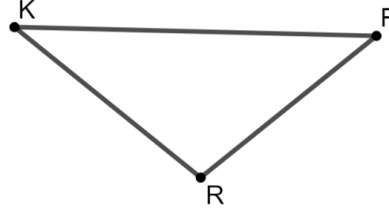
Angle	Measurement
$\angle L$	
$\angle P$	
$\angle R$	



7.H2.4 Guided Practice: Law of Cosines

1. $\triangle KFR$ is shown. $KF = 15$, $FR = 9$, and $KR = 10$. Find the missing angles.

Angle	Measurement
$\angle F$	
$\angle K$	
$\angle R$	



2. Two fishing boats begin traveling in the same direction from a port in Miami. One travels 20 nautical miles per hour on a bearing of 30° and the other travels 16 nautical miles per hour on a bearing of 45° . Assuming each boat maintains its course and speed, how far apart will the fishing boats be after three hours?



7.H2.5 Your Turn!

1. In $\triangle BNW$, $BN =$, $NW = 12$, and $BW = 13$. Find the missing angles.

Angle	Measurement
$\angle B$	
$\angle N$	
$\angle W$	

2. In $\triangle SLC$, $SC = 34$, $SL = 22$, and $m\angle LSC = 42^\circ$. Find all missing sides and angles.

Part	Measurement
$\overline{LC}$	
$\angle C$	
$\angle L$	



7.H2.6 Wrap-Up: Cool Down

1. Write a summary of what you learned today.

Unit 7, Lesson H3: Area of a Triangle Using the Sine Ratio



Benchmark

MA.912.T.1.4 Solve mathematical problems involving finding the area of a triangle given two sides and the included angle.



Learning Target

- ☐ I can find the area of a triangle using the sine ratio.



7.H3.1 Warm-Up: History of Mathematics

Hannah Fry is an English mathematician who works alongside physicists, mathematicians, computer scientists, architects, and geographers to study the patterns in human behavior. Fry's research applies to a wide range of social problems and questions, from shopping and transport to urban crime, riots, and terrorism.

Fry has also written two popular math books: *The Mathematics of Love: Patterns, Proofs, and the Search for the Ultimate Equation* and *The Indisputable Existence of Santa Claus*.

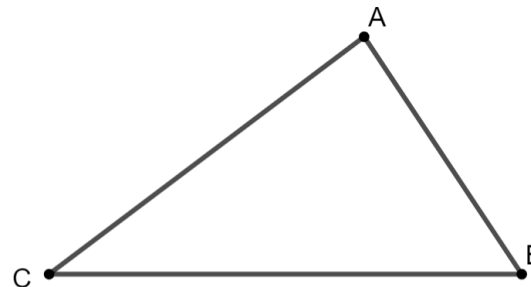
Fry is quoted as saying, "There is a mathematical angle to almost anything."

1. Interpret this quote in your own words.
2. What is a specific real-world example where mathematics is present?



7.H3.2 Exploration: Area of a Triangle

1. $\triangle ABC$ is shown.



- a. Use $\triangle ABC$ to do the following.
 - Draw an altitude from point A to side $\overline{CB}$
 - Label the altitude as h .
 - Label the point where the altitude intersects side $\overline{BC}$ as D .
 - Label the sides as $AB = c$, $BC = a$, and $AC = b$.

The altitude creates two right triangles inside $\triangle ABC$.

- b. Write the equation that can be used to find the area of $\triangle ABC$.
- c. Complete each equation using right triangle trigonometry.

$\triangle ADB$	$\triangle ADC$
$\sin(B) =$	$\sin(C) =$

d. Rewrite each equation in terms of h .

$\triangle ADB$	$\triangle ADC$
$h =$	$h =$

e. Substitute each expression into the formula,
 $A = \frac{1}{2}ah$.

$\triangle ADB$	$\triangle ADC$
$A = \frac{1}{2}a$ _____	$A = \frac{1}{2}a$ _____

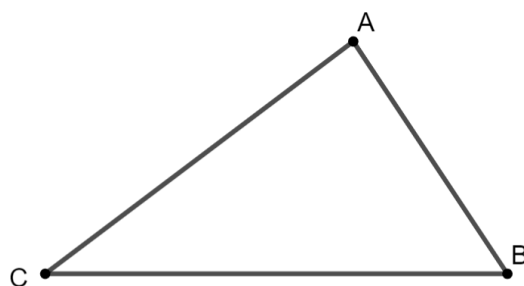
f. What do you notice about the two equations?

b. Find the area of the same triangle given that
 $m\angle A = 90^\circ$, $AC = 55$, and $AB = 48$.

c. Discuss with your partner why the formula $A = \frac{1}{2}bh$ could
also be used when given $m\angle A = 90^\circ$, $AC = 55$, and $AB = 48$.
Summarize your discussion.

2. $\triangle ABC$ is shown.

a. Find the area of
 $\triangle ABC$ given that
 $m\angle C = 41^\circ$,
 $AC = 55$, and
 $BC = 73$.





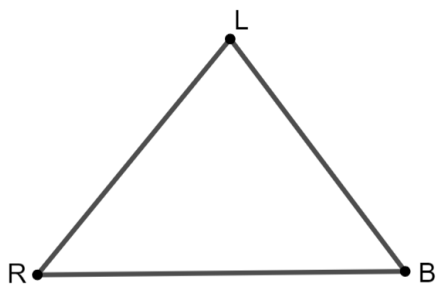
7.H3.3 Guided Practice: Area of a Triangle



The formulas for finding the area of a triangle using sine are:

$$A = \frac{1}{2}bc \sin A \quad A = \frac{1}{2}ac \sin B \quad A = \frac{1}{2}ab \sin C$$

1. $\triangle RLB$ is shown. $RL = 14$, $RB = 16.5$, and $m\angle LRB = 50^\circ$. Find the area of the triangle.



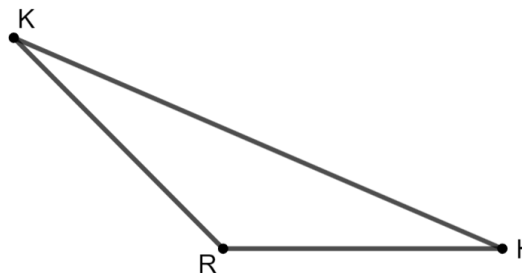
2. In $\triangle FPB$, $FP = 9$, $PB = 18$, and $m\angle FPB = 117^\circ$. Find the area of the triangle.



7.H3.4 Your Turn!

1. In $\triangle NBR$, $NB = 20$, $NR = 18.6$, and $m\angle RNB = 43^\circ$. Find the area of the triangle.

2. $\triangle KRH$ is shown. $KR = 18$, $RH = 16$, and $m\angle KRH = 135^\circ$. Find the area of the triangle.





7.H3.5 Wrap-Up: Cool Down

1. What are the similarities and differences between finding the area of a triangle and finding the area of a triangle using the sine ratio?

THIS PAGE WAS INTENTIONALLY LEFT BLANK.